

Chemistry Compendium

Chapters 1 · 2 · 3 · 4 · 7

Everything in five chapters: every definition, law, postulate, derivation, method, shape, structure, trend, standard value and formula — assembled in NCERT section order.

CONTENTS — EACH CHAPTER BEGINS ON A NEW PAGE

Ch 1 Some Basic Concepts of Chemistry

Matter and measurement · significant figures · the five laws of chemical combination · Dalton's theory · atomic and molecular masses · the mole concept · empirical and molecular formulae · stoichiometry and the limiting reagent · concentration of solutions

Ch 2 Structure of Atom

Discovery of the sub-atomic particles · Thomson and Rutherford models and their drawbacks · electromagnetic radiation · Planck's quantum · the photoelectric effect · atomic spectra and the Rydberg formula · Bohr's model and its limitations · de Broglie and Heisenberg · the quantum mechanical model, quantum numbers, orbital shapes, nodes and electronic configurations

Ch 3 Classification of Elements and Periodicity in Properties

Genesis of periodic classification · Moseley and the modern periodic law · structure of the table and period lengths · nomenclature above $Z = 100$ · the four blocks · metals, non-metals and metalloids · every periodic trend with its reason · the anomalies · chemical periodicity

Ch 4 Chemical Bonding and Molecular Structure

Lewis structures and formal charge · octet-rule failures · ionic bonding and lattice enthalpy · bond parameters · resonance · polarity and dipole moment · the complete VSEPR table with shapes · valence bond theory, sigma and pi bonds, hybridisation · molecular orbital theory with bond orders · hydrogen bonding

Ch 7 Redox Reactions

The three views of oxidation and reduction · oxidation number rules and fractional averages · types of redox change including disproportionation · both balancing methods worked in acidic and basic medium · redox titrations and n-factors · electrode processes and EMF

Ref Master Formula Sheet

Every formula, constant and standard value from all five chapters, with the VSEPR

How to read this. Each chapter follows NCERT's own section numbering, so a heading here matches the heading in your textbook. Boxes are colour-coded throughout: **grey** boxes carry the statement to memorise, **blue-edged** boxes are derivations or step-by-step methods, **green** boxes are worked examples, **yellow** boxes are exam notes, and **red** boxes are the traps that lose marks.

A FEW ITEMS GO BEYOND THE NCERT TEXT

These are standard exam formulae that NCERT states differently or omits. They are included because they get asked, but you can skip every one of them and still have complete NCERT coverage:

1. The **full Bohr derivation** from Coulomb's law (NCERT states r_n and E_n without deriving them), and the relations $PE = 2E$, $KE = -E$
2. $v_n = 2.188 \times 10^6 Z/n$ — NCERT gives radius and energy only
3. $\lambda = 12.27/\sqrt{V}$ Å and the shortcut $E(\text{eV}) = 1240/\lambda(\text{nm})$
4. Number of spectral lines $= n(n-1)/2$, and the stopping-potential form $eV_0 = h(\nu - \nu_0)$
5. **Isotones** (NCERT Chapter 2 covers isotopes and isobars only)
6. The **steric-number formula** $H = \frac{1}{2}[V + M - C + A]$ and **Fajans' rules** in Chapter 4
7. **Normality**, equivalent mass and $N_1V_1 = N_2V_2$ in Chapters 1 and 7

Some Basic Concepts of Chemistry

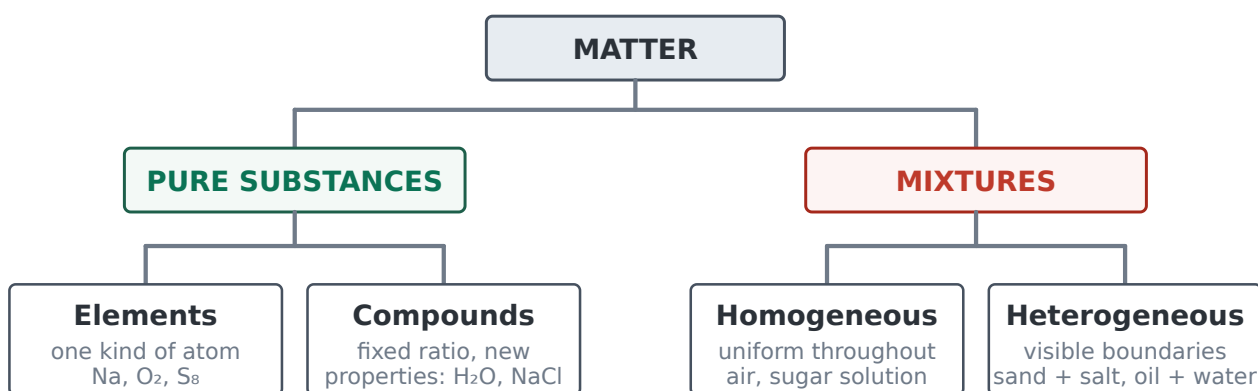
Complete chapter notes — every law, the full mole machinery, stoichiometry and solution concentration, with worked numericals and the traps that cost marks.

The spine of this chapter: chemistry counts particles but the laboratory can only weigh them. The **mole** is the bridge, and almost every numerical in this chapter — and in the whole two-year course — is the same three-step journey: **given quantity** → **moles** → **asked quantity**. Learn to spot which bridge (molar mass, Avogadro number, molar volume, molarity) each arrow needs and the chapter collapses to one idea.

1 • Matter, Properties and Measurement

1.1 Classification of matter

Matter is anything that has mass and occupies space. It is classified two ways — by **physical state** and by **chemical composition**.



Constituents of a mixture keep their own properties and can be separated by physical means;
constituents of a compound cannot — only a chemical reaction will break them apart.

Classification of matter by composition. The single most useful test in this diagram: can the parts be separated by a physical method (filtration, distillation, chromatography)? If yes it is a mixture; if it takes a chemical change, it is a compound.

1.2 Physical and chemical properties

Physical property	Chemical property
Measured without changing the identity of the substance	Observed only when the substance is converted into another substance
Colour, odour, melting point, boiling point, density, hardness	Combustibility, acidity, reactivity with water, rusting

1.3 SI units

Base quantity	Name	Symbol	Base quantity	Name	Symbol
Length	metre	m	Thermodynamic temperature	kelvin	K
Mass	kilogram	kg	Amount of substance	mole	mol
Time	second	s	Luminous intensity	candela	cd
Electric current	ampere	A			

Constants worth memorising: $1 \text{ L} = 1 \text{ dm}^3 = 10^3 \text{ cm}^3 = 10^{-3} \text{ m}^3$ · $1 \text{ bar} = 10^5 \text{ Pa}$ · $1 \text{ atm} = 1.013 \times 10^5 \text{ Pa} = 760 \text{ mm Hg}$ · density in SI is kg m^{-3} but in practice g cm^{-3} · $T(\text{K}) = t(^{\circ}\text{C}) + 273.15$ · $t(^{\circ}\text{F}) = \frac{9}{5} t(^{\circ}\text{C}) + 32$.

WHERE THIS LOSES MARKS

1. **Mass is not weight.** Mass is the quantity of matter (constant everywhere); weight is the force of gravity on it (varies with location).
2. The kelvin scale carries **no degree sign**: write 298 K, not 298°K. Negative kelvin temperatures do not exist.

2 · Handling Numbers

2.1 Precision and accuracy

Precision is how closely repeated measurements agree with *one another*; **accuracy** is how close a measurement is to the *true* value. A set of readings can be precise but inaccurate (a systematically mis-calibrated balance) — never assume one implies the other.

2.2 Significant figures

Significant figures are the digits known with certainty plus one final uncertain digit. They record how good a measurement is, so they must survive the arithmetic.

Rule	Example	Sig. figs
All non-zero digits are significant	285 cm	3
Zeros between non-zero digits are significant	2.005	4
Leading zeros are not significant — they only fix the decimal point	0.00234	3
Trailing zeros <i>after</i> a decimal point are significant	0.200	3
Trailing zeros in a whole number with <i>no</i> decimal point are ambiguous — use scientific notation	$100 \rightarrow 1.0 \times 10^2$	2
Exact numbers (counting, defined conversions) have infinite sig. figs	1 dozen = 12; 1 m = 100 cm	∞

Arithmetic rules. Addition/subtraction: the answer keeps the fewest *decimal places*.
Multiplication/division: the answer keeps the fewest *significant figures*.

MEMORY HOOK

Addition → **A**fter the point (decimal places). Multiplication → count the **M**eaningful digits (significant figures).

WORKED EXAMPLE 1 — SIGNIFICANT FIGURES IN ARITHMETIC

(a) $12.11 + 18.0 + 1.012 = 31.122 \Rightarrow \mathbf{31.1}$ (18.0 has only one decimal place).

(b) $2.5 \times 1.25 = 3.125 \Rightarrow \mathbf{3.1}$ (2.5 has only two significant figures).

Rounding off. If the digit to be dropped is greater than 5, raise the preceding digit by one; if less than 5, leave it; if it is exactly 5, raise the preceding digit only when it is **odd** (so 6.35 → 6.4, but 6.25 → 6.2).

2.3 Dimensional analysis (the factor-label method)

Multiply the given quantity by conversion factors written as fractions, arranged so that every unwanted unit cancels. The method never asks you to remember whether to multiply or divide — the units decide.

WORKED EXAMPLE 2 — UNIT CONVERSION

Convert 2.0 L to m^3 : $2.0 \text{ L} \times \frac{10^{-3} \text{ m}^3}{1 \text{ L}} = \mathbf{2.0 \times 10^{-3} \text{ m}^3}$.

Convert a density of 1.15 g cm^{-3} to kg m^{-3} : $\frac{1.15 \text{ g}}{\text{cm}^3} \times \frac{1 \text{ kg}}{10^3 \text{ g}} \times \frac{10^6 \text{ cm}^3}{1 \text{ m}^3} = \mathbf{1.15 \times 10^3 \text{ kg m}^{-3}}$.

3 · Laws of Chemical Combination

Five experimental laws were established before any atomic theory existed. Dalton's theory was invented afterwards, to explain them.

3.1 Law of conservation of mass (Lavoisier, 1789)

Matter can neither be created nor destroyed in a chemical reaction: **total mass of reactants = total mass of products**.

This is why chemical equations must be balanced, and it is the licence for every stoichiometric calculation you will ever do.

3.2 Law of definite (constant) proportions (Proust, 1799)

A given compound always contains exactly the **same elements in the same proportion by mass**, whatever its source or method of preparation.

Pure water from a river, from combustion of hydrogen, or from a laboratory synthesis is always 1 : 8 hydrogen to oxygen by mass. This law is what makes a chemical formula meaningful at all.

3.3 Law of multiple proportions (Dalton, 1803)

If two elements combine to form more than one compound, the masses of one element that combine with a **fixed mass** of the other are in a **simple whole-number ratio**.

WORKED EXAMPLE 3 — VERIFYING THE LAW OF MULTIPLE PROPORTIONS

Carbon and oxygen form CO and CO₂. Fix the carbon at 12 g:

In CO, 12 g C combines with **16 g O**. In CO₂, 12 g C combines with **32 g O**.

Ratio of the oxygen masses = 16 : 32 = **1 : 2**, a simple whole-number ratio. The law is obeyed.

Same treatment for H₂O and H₂O₂: fix 2 g of hydrogen; oxygen masses are 16 g and 32 g, again 1 : 2.

3.4 Gay-Lussac's law of gaseous volumes (1808)

When gases combine or are produced in a chemical reaction, they do so in a **simple whole-number ratio by volume**, provided all volumes are measured at the same temperature and pressure.

$\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$ combines in the volume ratio 1 : 1 : 2 — exactly the ratio of the coefficients.

3.5 Avogadro's law (1811)

Equal volumes of all gases, at the same temperature and pressure, contain equal numbers of molecules.

Avogadro's insight was that the particles of an element need not be single atoms: taking hydrogen and oxygen to be *diatomic* resolved the contradiction between Gay-Lussac's volume ratios and Dalton's atomic theory. Its immediate consequence is that for gases, **volume ratio = mole ratio** — the reason gaseous stoichiometry can skip masses entirely.

FREQUENTLY ASKED

"Which law explains why gaseous reaction volumes are in simple ratios?" — Gay-Lussac's law is the *observation*; Avogadro's law is the *explanation*. Mention both, in that order.

4 · Dalton's Atomic Theory (1808)

1. Matter consists of indivisible atoms.
2. All atoms of a given element are identical in mass and properties; atoms of different elements differ.
3. Compounds form when atoms of different elements combine in fixed, simple whole-number ratios.
4. Chemical reactions only rearrange atoms — atoms are neither created nor destroyed.

What it explained: postulate 4 gives the law of conservation of mass; postulate 3 gives the laws of definite and multiple proportions.

LIMITATIONS YOU MUST BE ABLE TO STATE

1. It could not explain the law of gaseous volumes (that needed Avogadro's molecules).
2. Atoms are **not** indivisible — they contain electrons, protons and neutrons.
3. Atoms of the same element are **not** always identical in mass — **isotopes** exist (and isobars show different elements can share a mass).
4. It cannot explain why atoms bond at all, nor the existence of non-stoichiometric compounds.

5 · Atomic and Molecular Masses

5.1 The unified mass unit

1 u (unified mass unit, also called 1 dalton, Da) = $\frac{1}{12}$ of the mass of one atom of ^{12}C = 1.66056×10^{-24} g.

On this scale the ^{12}C atom is exactly 12 u *by definition*, hydrogen is 1.008 u and oxygen 16.00 u.

5.2 Average atomic mass

Because elements occur as mixtures of isotopes, the atomic mass in the periodic table is a **weighted average** over natural abundance:

$$\text{Average atomic mass} = \sum (\text{fractional abundance} \times \text{isotopic mass})$$

WORKED EXAMPLE 4 — AVERAGE ATOMIC MASS OF CHLORINE

Chlorine is 75.77% ^{35}Cl (34.9689 u) and 24.23% ^{37}Cl (36.9659 u).

$$\bar{A} = (0.7577)(34.9689) + (0.2423)(36.9659) = 26.4959 + 8.9568 = \mathbf{35.45 \text{ u}}$$

Notice the answer lies nearer 35 than 37 — the weighted average always leans towards the more abundant isotope. That single observation is enough to eliminate wrong options in an MCQ.

5.3 Molecular mass and formula mass

Molecular mass is the sum of the atomic masses of all atoms in a molecule: for CH_4 , $12.011 + 4(1.008) = 16.043$ u.

Formula mass is used for ionic compounds, which have no discrete molecules — NaCl is a continuous lattice, so we quote the formula mass $22.99 + 35.45 = 58.44$ u for the formula unit NaCl .

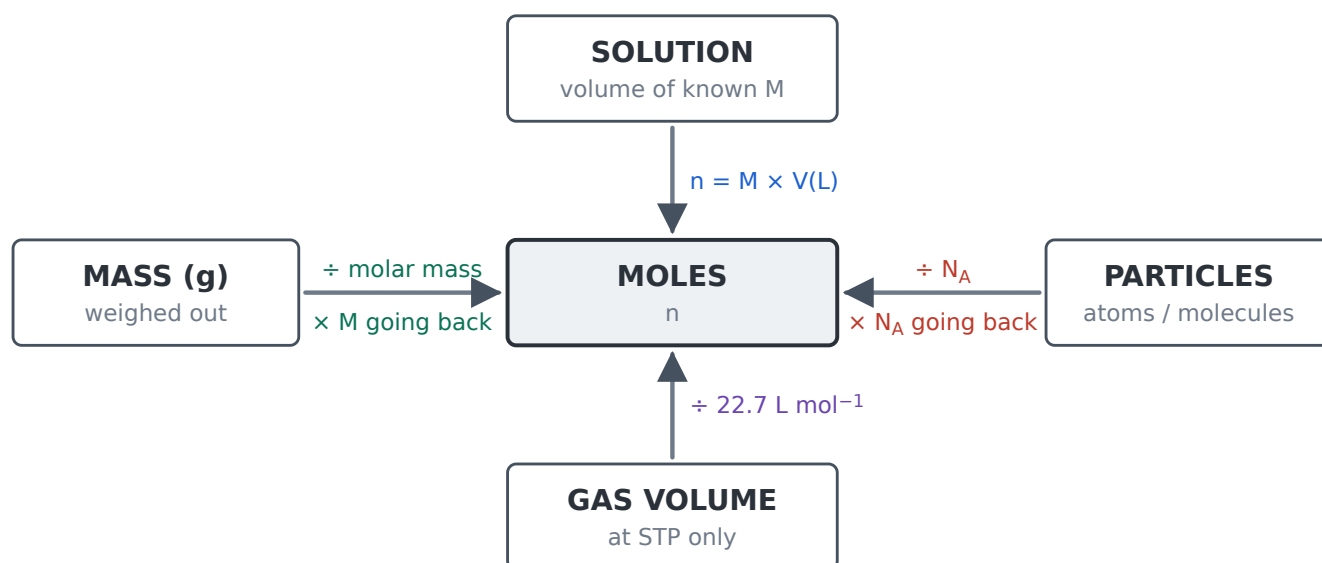
WHERE THIS LOSES MARKS

1. Writing "molecular mass of NaCl ". Ionic solids have a **formula mass**; there is no NaCl molecule.
2. Atomic/molecular mass is in **u** (a mass of one particle); molar mass is in **g mol^{-1}** (a mass of one mole). They are numerically equal, which is a deliberate design of the scale — not a coincidence you may ignore in the units.

6 · The Mole Concept

One **mole** is the amount of substance containing exactly 6.022×10^{23} elementary entities. $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$ is the **Avogadro constant**.

The entity must always be named: one mole of O means 6.022×10^{23} oxygen *atoms*; one mole of O_2 means that many *molecules*, and therefore $2N_A$ atoms.



The mole map. Every numerical in this chapter is a path through this diagram. Reverse an arrow and the operation inverts: mass $\rightarrow$ mole divides by M , mole $\rightarrow$ mass multiplies by M .

6.1 Molar mass

$n = \frac{m}{M}$ where m is mass in g and M is molar mass in g mol^{-1} . Also $n = \frac{N}{N_A}$ and, for a gas at STP, $n = \frac{V}{22.7 \text{ L}}$.

6.2 Molar volume and the STP trap

22.7 L OR 22.4 L?

1. **STP is now 273.15 K and 1 bar**, at which the molar volume of an ideal gas is **22.7 L mol^{-1}** (22.71098 L). This is the value the current NCERT text uses.
2. The older **22.4 L mol^{-1}** applies at 273.15 K and **1 atm**. It is not wrong — it answers a different question.
3. Read the pressure stated in the question. If it says 1 bar use 22.7; if it says 1 atm use 22.4; state which you used.
4. Molar volume applies to **gases only**. There is no 22.7 L of liquid water.

WORKED EXAMPLE 5 — THE STANDARD MOLE CONVERSIONS

- (a) **Moles in 5.6 g of N₂:** $M = 28.0 \text{ g mol}^{-1}$, so $n = 5.6/28.0 = \mathbf{0.20 \text{ mol}}$.
- (b) **Molecules present:** $N = nN_A = 0.20 \times 6.022 \times 10^{23} = 1.2 \times 10^{23} \text{ molecules}$.
- (c) **Atoms present:** N₂ is diatomic, so $2 \times 1.2 \times 10^{23} = 2.4 \times 10^{23} \text{ atoms}$.
- (d) **Volume at STP:** $0.20 \times 22.7 = \mathbf{4.54 \text{ L}}$.

WORKED EXAMPLE 6 — COUNTING ATOMS INSIDE A COMPOUND

How many oxygen atoms are in 9.8 g of H₂SO₄?

$M(\text{H}_2\text{SO}_4) = 2(1.008) + 32.06 + 4(16.00) = 98.08 \approx 98 \text{ g mol}^{-1}$, so $n = 9.8/98 = 0.10 \text{ mol}$.

Each formula unit carries 4 oxygen atoms $\Rightarrow$ moles of O atoms = 0.40 mol.

$N = 0.40 \times 6.022 \times 10^{23} = 2.41 \times 10^{23} \text{ oxygen atoms}$.

7 · Percentage Composition and Formulae

7.1 Mass percentage of an element

$$\% \text{ element} = \frac{\text{total mass of that element in one mole of compound}}{\text{molar mass of the compound}} \times 100$$

For H₂O: $\% \text{H} = \frac{2 \times 1.008}{18.02} \times 100 = 11.19\%$ and $\% \text{O} = \frac{16.00}{18.02} \times 100 = 88.79\%$.

7.2 Empirical and molecular formula

Empirical formula	Molecular formula
The simplest whole-number ratio of atoms	The actual number of atoms in one molecule
CH for benzene; HO for hydrogen peroxide; CH ₂ O for glucose	C ₆ H ₆ ; H ₂ O ₂ ; C ₆ H ₁₂ O ₆

Molecular formula = (empirical formula) _{n} where $n = \frac{\text{molar mass}}{\text{empirical formula mass}}$. If $n = 1$ the two formulae coincide (H₂O, NH₃, CO₂).

METHOD 1 Percentage composition → molecular formula

1. Take **100 g** of the compound, so each percentage becomes a mass in grams.
2. Divide each mass by the atomic mass of that element to get **moles of atoms**.
3. Divide every result by the **smallest** of them — this gives the simplest ratio.
4. If any ratio is still fractional, multiply *all* of them by the smallest integer that clears it (×2 for .5, ×3 for .33, ×4 for .25). Write the empirical formula.
5. Compute the empirical formula mass, find $n = M/\text{EFM}$, and multiply every subscript by n .

The steps never change; only the arithmetic does. Step 4 is where most answers go wrong — never round 1.5 to 2.

WORKED EXAMPLE 7 — THE STANDARD NCERT NUMERICAL

A compound is 4.07% H, 24.27% C, 71.65% Cl by mass and has molar mass 98.96 g mol⁻¹. Find both formulae.

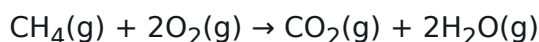
Element	Mass in 100 g	Moles	÷ smallest (2.021)
H	4.07 g	4.07/1.008 = 4.04	2.00
C	24.27 g	24.27/12.01 = 2.021	1.00
Cl	71.65 g	71.65/35.45 = 2.021	1.00

Empirical formula = **CH₂Cl**; empirical formula mass = 12.01 + 2(1.008) + 35.45 = 49.48.

$n = 98.96/49.48 = 2 \Rightarrow$ molecular formula = **C₂H₄Cl₂**.

8 · Stoichiometry and the Limiting Reagent

A balanced equation is read as a **mole statement**, never a mass statement:



reads as "1 mole of methane reacts with 2 moles of dioxygen" — and, since all four are gases, also "1 volume with 2 volumes" by Avogadro's law. It does **not** read as 1 g with 2 g.

8.1 The limiting reagent

The **limiting reagent** is the reactant that is completely consumed first; it fixes the maximum amount of product. Every other reactant is in **excess** and is left over.

METHOD 2 Identifying the limiting reagent — the division test

1. Convert the given mass (or volume) of **each** reactant to moles.
2. Divide each reactant's moles by **its own stoichiometric coefficient** in the balanced equation.
3. The **smallest** quotient identifies the limiting reagent.
4. Base *all* product calculations on the limiting reagent alone.
5. Excess left over = (moles supplied) – (moles actually consumed, from the limiting reagent's ratio).

Never decide the limiting reagent by comparing raw masses, or even raw moles — only moles divided by coefficients are comparable.

WORKED EXAMPLE 8 — LIMITING REAGENT IN THE HABER PROCESS

50.0 kg of N_2 and 10.0 kg of H_2 are mixed: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$. Find the mass of NH_3 formed.

$$n(\text{N}_2) = \frac{50.0 \times 10^3}{28.0} = 1786 \text{ mol} \rightarrow 1786/1 = 1786$$

$$n(\text{H}_2) = \frac{10.0 \times 10^3}{2.016} = 4960 \text{ mol} \rightarrow 4960/3 = 1653 \leftarrow \text{smaller, so H}_2 \text{ is limiting}$$

$$n(\text{NH}_3) = \frac{2}{3} \times 4960 = 3307 \text{ mol}; m = 3307 \times 17.03 = 5.63 \times 10^4 \text{ g} = \mathbf{56.3 \text{ kg of NH}_3}$$

N_2 consumed = $4960/3 = 1653 \text{ mol}$, so N_2 left over = $1786 - 1653 = 133 \text{ mol} = \mathbf{3.7 \text{ kg unreacted N}_2}$.

Note how badly the raw-mass instinct fails here: there is five times as much nitrogen by mass, and nitrogen is still the reactant in excess.

8.2 Percentage yield

$\% \text{ yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100$, where the theoretical yield is what the limiting reagent predicts.

9 · Concentration of Solutions

Throughout: the **solute** is the dissolved substance, the **solvent** is the medium, and mass of solution = mass of solute + mass of solvent.

Term	Definition	Formula	Depends on T ?
Mass percentage	Mass of solute per 100 g of solution	$\frac{w_B}{w_A + w_B} \times 100$	No
Mole fraction x_B	Moles of a component per total moles	$\frac{n_B}{n_A + n_B}$; $x_A + x_B = 1$	No
Molarity (M)	Moles of solute per litre of solution	$M = \frac{n_B}{V_{\text{solution}}(\text{L})}$	Yes
Molality (m)	Moles of solute per kilogram of solvent	$m = \frac{n_B}{w_A(\text{kg})}$	No
ppm	Parts of solute per 10^6 parts of solution	$\frac{\text{mass of solute}}{\text{mass of solution}} \times 10^6$	No

THE ONE-LINE REASON

Molarity is temperature-dependent because it is defined per unit **volume**, and volume expands on heating; molality is defined per unit **mass** of solvent, and mass does not change with temperature. Expect this as a one-mark question — and it is the reason molality is preferred in colligative-property work next year.

DERIVATION 1 Molarity from mass percentage and density

1. Take exactly **1 L = 1000 mL** of solution. Its mass is $1000 d$ g, where d is the density in g mL^{-1} .
2. If the solution is $x\%$ by mass, the mass of solute in it is $\frac{x}{100} \times 1000 d = 10 x d$ g.
3. Moles of solute = $\frac{10 x d}{M_B}$, where M_B is the molar mass of the solute.
4. This is contained in exactly 1 L, so it *is* the molarity.

$$M = \frac{10 \times (\text{mass } \%) \times d}{M_B} \text{ — with } d \text{ in } \text{g mL}^{-1}. \text{ The 10 comes from } 1000 \text{ mL} \div 100\%.$$

DERIVATION 2 Converting molarity to molality

1. Again take 1 L of solution: mass of solution = $1000 d$ g and moles of solute = M (numerically, by definition of molarity).
2. Mass of solute = $M \times M_B$ g.
3. Mass of **solvent** = mass of solution – mass of solute = $(1000 d - M M_B)$ g = $\frac{1000d - M M_B}{1000}$ kg.
4. Molality = moles of solute per kg of solvent = $\frac{M}{(1000d - M M_B)/1000}$.

$m = \frac{1000 M}{1000 d - M M_B}$. For dilute aqueous solutions $M M_B \ll 1000d$ and $d \approx 1$, so $m \approx M$ — which is why the two are often confused.

9.1 Dilution

Adding solvent changes the volume but not the moles of solute: $M_1 V_1 = M_2 V_2$. For mixing two solutions of the same solute: $M_{\text{final}} = \frac{M_1 V_1 + M_2 V_2}{V_1 + V_2}$.

WORKED EXAMPLE 9 — CONCENTRATED ACID TO MOLARITY

Commercial concentrated HCl is 38% by mass with density 1.19 g mL^{-1} . Find its molarity, and the volume needed to prepare 250 mL of 0.50 M HCl.

$$M = \frac{10 \times 38 \times 1.19}{36.46} = \mathbf{12.4 \text{ M}}$$

Dilution: $M_1 V_1 = M_2 V_2 \Rightarrow V_1 = \frac{0.50 \times 250}{12.4} = \mathbf{10.1 \text{ mL of the concentrated acid}}$, made up to 250 mL.

WORKED EXAMPLE 10 — MOLALITY AND MOLE FRACTION TOGETHER

2.5 g of ethanoic acid (CH_3COOH , $M = 60 \text{ g mol}^{-1}$) is dissolved in 75 g of benzene ($M = 78 \text{ g mol}^{-1}$). Find the molality and the mole fraction of the acid.

$$n_B = 2.5/60 = 0.0417 \text{ mol}; \quad n_A = 75/78 = 0.962 \text{ mol}.$$

$$m = \frac{0.0417}{0.075 \text{ kg}} = \mathbf{0.56 \text{ mol kg}^{-1}} \quad x_B = \frac{0.0417}{0.0417 + 0.962} = \mathbf{0.0416}$$

10 · Where marks are lost

THE TEN TRAPS OF CHAPTER 1

1. **Molar volume is 22.7 L at 1 bar, 22.4 L at 1 atm** — and applies to gases only. Say which standard you are using.
2. **Molarity is temperature-dependent; molality is not.** Molarity uses volume of *solution*, not of solvent.
3. **Never decide a limiting reagent from masses.** Moles $\div$ coefficient, then compare.
4. **Leading zeros are never significant; trailing zeros after a decimal point always are.** 0.0500 has three significant figures.
5. **Addition follows decimal places, multiplication follows significant figures.** Mixing the two rules is the commonest silent error.
6. **Do not round an empirical-formula ratio of 1.5 up to 2** — multiply every ratio by 2 instead.
7. **One mole of O_2 contains $2N_A$ atoms** but N_A molecules. Always ask what the question is counting.
8. **Average atomic mass is weighted by abundance**, never a plain mean of the isotopic masses.
9. **Ionic compounds have formula mass, not molecular mass.**
10. **Mass of solution = solute + solvent.** "75 g of benzene" is the solvent mass, so the solution weighs 77.5 g — a favourite one-mark deduction in molality problems.

Chapter 1 of the rationalised NCERT (2023-24 onward). Atomic masses used throughout: H 1.008, C 12.01, N 14.01, O 16.00, Cl 35.45, S 32.06, Na 22.99 u.

Structure of Atom

Complete chapter notes — every model and why it failed, all the standard derivations worked in full, the quantum numbers, and the configuration rules with their exceptions.

The spine of this chapter: it is one long argument, not a list of facts. Each model is destroyed by one experiment and replaced by the next — Thomson by **α -scattering**, Rutherford by **classical radiation loss and line spectra**, Bohr by **many-electron atoms and the uncertainty principle**. Learn the chain "model → the one fact it cannot explain → successor" and half the theory questions answer themselves.

1 · Sub-atomic Particles

1.1 Discovery of the electron — cathode rays

In a discharge tube at very low pressure (10^{-4} atm) and high voltage, rays stream from the cathode to the anode.

- They travel in straight lines and cast sharp shadows.
- They are **deflected towards the positive plate** in an electric field $\Rightarrow$ they carry **negative** charge.
- Their properties (charge-to-mass ratio) are **independent of the gas and of the electrode material** $\Rightarrow$ the electron is a universal constituent of all matter.

Thomson (1897) measured the charge-to-mass ratio by balancing electric and magnetic deflections: $\frac{e}{m_e} = 1.758820 \times 10^{11} \text{ C kg}^{-1}$.

Millikan's oil-drop experiment (1909) measured the charge itself, $e = 1.602 \times 10^{-19} \text{ C}$, and showed that charge is always an integral multiple of e — charge is **quantised**. Combining the two gives $m_e = \frac{1.602 \times 10^{-19}}{1.759 \times 10^{11}} = 9.1094 \times 10^{-31} \text{ kg}$.

1.2 Protons and neutrons

Anode (canal) rays, observed with a perforated cathode, are positively charged. Unlike cathode rays, their charge-to-mass ratio **depends on the gas** taken, and it is largest for hydrogen — giving the lightest positive particle, the **proton**. **Chadwick (1932)** bombarded beryllium with α -particles and found an electrically neutral particle of nearly the proton's mass: the **neutron**.

Particle	Symbol	Absolute charge (C)	Relative charge	Mass (kg)	Mass (u)
Electron	e^-	-1.602×10^{-19}	-1	9.1094×10^{-31}	0.00055
Proton	p^+	$+1.602 \times 10^{-19}$	+1	1.67262×10^{-27}	1.00727
Neutron	n	0	0	1.67493×10^{-27}	1.00867

A proton is about **1837 times** heavier than an electron — which is why essentially all of an atom's mass sits in the nucleus.

1.3 Atomic number, mass number, isotopes and isobars

Atomic number Z = number of protons = number of electrons in a neutral atom. Mass number $A = Z +$ number of neutrons. Written as ${}_Z^AX$.

Term	Same	Different	Example
Isotopes	Z (same element)	A (neutron number)	${}_1^1\text{H}, {}_1^2\text{H}, {}_1^3\text{H} \cdot {}_{17}^{35}\text{Cl}, {}_{17}^{37}\text{Cl}$
Isobars	A	Z (different elements)	${}_{6}^{14}\text{C}$ and ${}_{7}^{14}\text{N}$
Isotones	Number of neutrons	Z and A	${}_{6}^{14}\text{C}$ and ${}_{8}^{16}\text{O}$ (8 neutrons each)
Isoelectronic	Number of electrons	Nuclear charge	$\text{N}^{3-}, \text{O}^{2-}, \text{F}^-, \text{Ne}, \text{Na}^+, \text{Mg}^{2+}$

FREQUENTLY ASKED

Chemical properties are governed by the **electrons**, so isotopes of an element are chemically almost identical and differ only in physical properties that depend on mass (density, rate of diffusion, rate of reaction in kinetic-isotope situations).

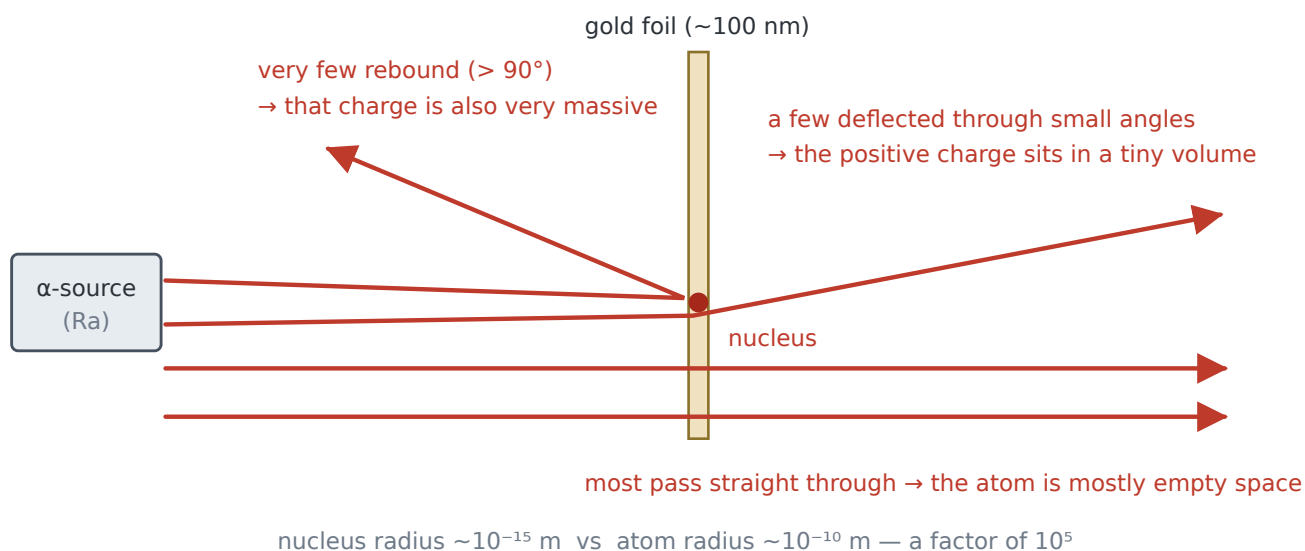
2 · Atomic Models and Their Failures

2.1 Thomson's model (1898)

The atom as a uniform sphere of positive charge with electrons embedded in it — the "plum pudding" or "watermelon" model. It accounted for electrical neutrality and nothing else, and was demolished by α -scattering.

2.2 Rutherford's α -scattering experiment (1911)

A thin gold foil (~ 100 nm) was bombarded with fast-moving α -particles (He^{2+}) from a radioactive source, detected on a fluorescent ZnS screen.



The α -scattering experiment. Each observation forces one conclusion, and the exam expects them paired: passes through → empty space; small deflection → positive charge concentrated in a tiny volume; rebound → that volume also holds nearly all the mass.

Rutherford's nuclear model. The positive charge and nearly all the mass are concentrated in a minute **nucleus**; electrons revolve around it at high speed in circular orbits; the electrostatic attraction supplies the centripetal force.

THE TWO FATAL DRAWBACKS — LEARN BOTH

1. **Stability.** By classical electrodynamics an *accelerating* charged particle continuously radiates energy. An orbiting electron accelerates, so it should spiral into the nucleus within about 10^{-8} s. Matter would not exist.
2. **Spectra.** A continuously spiralling electron would emit a **continuous** spectrum. Atoms actually give sharp **line** spectra.

It also said nothing about how the electrons are arranged, and so nothing about chemical behaviour.

3 · The Quantum Ideas That Rescued the Atom

3.1 Wave nature of electromagnetic radiation

$c = \nu\lambda$ with $c = 3.0 \times 10^8 \text{ m s}^{-1}$. Wavenumber $\bar{\nu} = \frac{1}{\lambda}$ (cm^{-1}), the number of waves per unit length.

Order of the electromagnetic spectrum by **increasing** frequency (decreasing wavelength): radio < microwave < infrared < **visible** < ultraviolet < X-rays < γ-rays. Visible light runs from about 400 nm (violet) to 750 nm (red).

3.2 Planck's quantum theory (1900)

Classical physics predicted that a black body would radiate ever more strongly at short wavelengths (the "ultraviolet catastrophe"); the observed curve instead peaks and falls. Planck resolved it by abandoning continuous energy:

Energy is emitted or absorbed **only in discrete packets** called quanta, of energy $E = h\nu = \frac{hc}{\lambda}$, with $h = 6.626 \times 10^{-34}$ J s. Emitted energy is always an integral multiple, $n h \nu$.

For light, one quantum is called a **photon**.

3.3 The photoelectric effect (Hertz 1887; Einstein 1905)

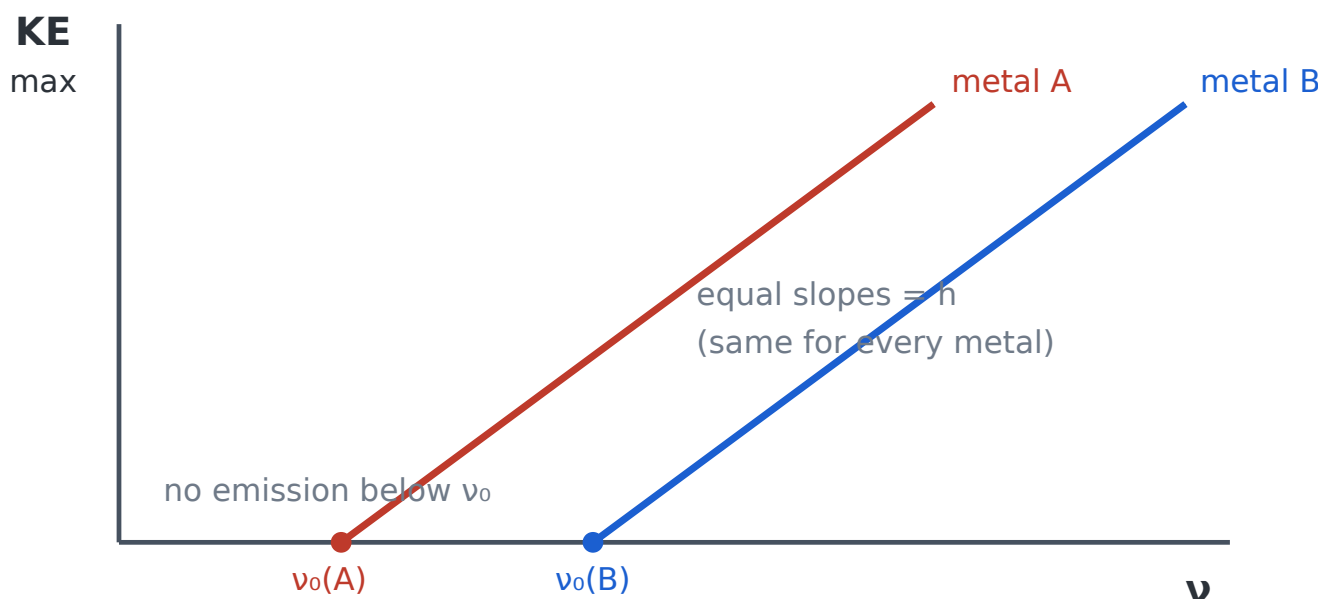
Electrons are ejected from a metal surface when light of sufficiently high frequency falls on it. Four observations, none of which the wave theory can explain:

Observation	Wave theory predicts	Photon theory explains
Below a threshold frequency ν_0 no electron is ejected, however intense the light	Enough intensity should always work	One photon, one electron: if $h\nu < h\nu_0$ the single photon simply cannot pay the price
Emission is instantaneous ($< 10^{-9}$ s)	A time lag while energy accumulates	The transfer is a single collision
Kinetic energy of the electrons rises with frequency , not with intensity	KE should rise with intensity	$E = h\nu$ fixes the energy per photon
Number of electrons (photocurrent) rises with intensity	—	Intensity = number of photons per second

DERIVATION 1 Einstein's photoelectric equation

1. A single photon of energy $h\nu$ is absorbed by a single electron.
2. Part of that energy, the **work function** $\phi = h\nu_0$, is spent freeing the electron from the metal surface.
3. Energy conservation gives the rest to the electron as kinetic energy:
4. $h\nu = h\nu_0 + \frac{1}{2}m_e v^2$.
5. If the electron is stopped by a retarding potential V_0 , then $\frac{1}{2}m_e v^2 = eV_0$, so $eV_0 = h(\nu - \nu_0)$.

$KE_{\max} = h\nu - h\nu_0 = h(\nu - \nu_0)$. **A graph of KE against ν is a straight line of slope h with x -intercept ν_0 — the same slope for every metal, since h is universal.**



Photoelectric graph. Two metals give **parallel** lines: the slope is Planck's constant and cannot change, while the intercept ν_0 measures the work function. Metal B, further right, has the larger work function.

WORKED EXAMPLE 1 — PHOTOELECTRIC NUMERICAL

The work function of caesium is 1.9 eV. Find its threshold frequency and the kinetic energy of an electron ejected by light of wavelength 500 nm.

$$\phi = 1.9 \times 1.602 \times 10^{-19} = 3.04 \times 10^{-19} \text{ J} \Rightarrow \nu_0 = \frac{\phi}{h} = \frac{3.04 \times 10^{-19}}{6.626 \times 10^{-34}} = 4.59 \times 10^{14} \text{ Hz}$$

$$E = \frac{hc}{\lambda} = \frac{(6.626 \times 10^{-34})(3 \times 10^8)}{500 \times 10^{-9}} = 3.98 \times 10^{-19} \text{ J}$$

$$KE = 3.98 \times 10^{-19} - 3.04 \times 10^{-19} = 9.4 \times 10^{-20} \text{ J} = 0.59 \text{ eV}$$

3.4 Atomic spectra and the hydrogen spectrum

A **continuous** spectrum (white light through a prism) has no gaps. An **emission line spectrum** is a set of bright lines on a dark background — the fingerprint of the element. An **absorption spectrum** is the exact negative: dark lines at the same positions in a continuous background.

Rydberg's formula for hydrogen: $\bar{\nu} = \frac{1}{\lambda} = 109,677 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ cm}^{-1}$, with $n_2 > n_1$.

Series	n_1	n_2	Region
Lyman	1	2, 3, 4 ...	Ultraviolet
Balmer	2	3, 4, 5 ...	Visible
Paschen	3	4, 5, 6 ...	Infrared
Brackett	4	5, 6, 7 ...	Infrared
Pfund	5	6, 7, 8 ...	Far infrared

MEMORY HOOK

Lyman **B**almer **P**aschen **B**rackett **P**fund → "Light **B**ulbs **P**roduce **B**right **P**hotons", with $n_1 = 1, 2, 3, 4, 5$ in that order. Only **Balmer** lies in the visible region.

4 · Bohr's Model of the Hydrogen Atom (1913)

4.1 Postulates

1. The electron revolves around the nucleus in certain **fixed circular orbits** of definite energy, called stationary states or energy shells.
2. While in a stationary state the electron **does not radiate** energy — this is a postulate, introduced precisely to defeat the classical objection.
3. Energy is emitted or absorbed **only when the electron jumps** between stationary states, with $\Delta E = E_2 - E_1 = h\nu$ (Bohr's frequency rule).
4. The angular momentum of the electron is **quantised**: $mvr = n\frac{h}{2\pi}$, $n = 1, 2, 3, \dots$

DERIVATION 2 Radius and energy of the n th Bohr orbit

Part A — radius.

1. The electrostatic attraction between a nucleus of charge $+Ze$ and the electron supplies the centripetal force: $\frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r^2} = \frac{mv^2}{r} \Rightarrow mv^2 = \frac{Ze^2}{4\pi\epsilon_0 r}$... (i)
2. Quantisation of angular momentum: $mvr = \frac{nh}{2\pi} \Rightarrow v = \frac{nh}{2\pi mr}$... (ii)
3. Substitute (ii) into (i): $m \left(\frac{nh}{2\pi mr} \right)^2 = \frac{Ze^2}{4\pi\epsilon_0 r}$
4. $\frac{n^2 h^2}{4\pi^2 m r^2} = \frac{Ze^2}{4\pi\epsilon_0 r} \Rightarrow r = \frac{n^2 h^2 \epsilon_0}{\pi m Z e^2}$

$$r_n = 52.9 \frac{n^2}{Z} \text{ pm} = 0.529 \frac{n^2}{Z} \text{ \AA} \text{ — so } r \propto n^2 \text{ and } r \propto 1/Z.$$

Part B — energy.

1. Kinetic energy: from (i), $\text{KE} = \frac{1}{2}mv^2 = \frac{Ze^2}{8\pi\epsilon_0 r}$.
2. Potential energy of the pair: $\text{PE} = -\frac{Ze^2}{4\pi\epsilon_0 r}$ (negative, since the force is attractive).
3. Total energy $E = \text{KE} + \text{PE} = \frac{Ze^2}{8\pi\epsilon_0 r} - \frac{Ze^2}{4\pi\epsilon_0 r} = -\frac{Ze^2}{8\pi\epsilon_0 r}$.
4. Insert the radius from Part A: $E_n = -\frac{mZ^2 e^4}{8\epsilon_0^2 h^2 n^2}$.

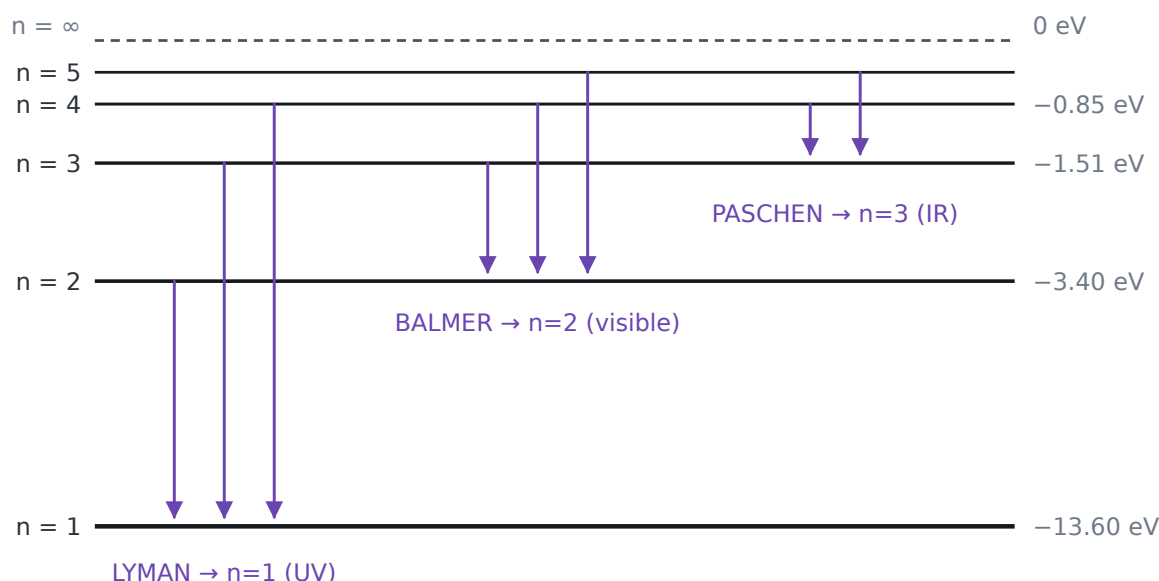
$$E_n = -2.18 \times 10^{-18} \frac{Z^2}{n^2} \text{ J} = -13.6 \frac{Z^2}{n^2} \text{ eV. Note PE} = 2E \text{ and KE} = -E \text{ — so the total energy is negative, which is exactly what "bound" means.}$$

$$\text{Also worth carrying: } v_n = 2.188 \times 10^6 \frac{Z}{n} \text{ m s}^{-1}, \text{ so } v \propto Z/n.$$

DERIVATION 3 The Rydberg formula from Bohr's model

1. An electron falls from level n_2 to level n_1 , emitting one photon: $\Delta E = E_{n_2} - E_{n_1}$.
2. $\Delta E = -2.18 \times 10^{-18} \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right) = 2.18 \times 10^{-18} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ J.}$
3. By Bohr's frequency rule $\Delta E = h\nu$, so $\nu = \frac{\Delta E}{h} = 3.29 \times 10^{15} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ Hz.}$
4. Wavenumber $\bar{\nu} = \frac{\nu}{c} = \frac{\Delta E}{hc}$.

$$\bar{\nu} = 109,677 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ cm}^{-1} \text{ — Rydberg's empirical formula, now derived, with } R_H \text{ obtained from } m, e, h \text{ and } \epsilon_0 \text{ alone. This agreement is the reason Bohr's model was accepted.}$$



the levels crowd together as n grows, because $E \propto -1/n^2$ — so every series converges to a limit

Hydrogen energy levels (vertical scale compressed above $n = 3$ so the upper levels stay legible). The spacing between consecutive levels decreases as n rises, so each series has a convergence limit. The series is named by where the electron **lands** (n_1), not where it starts.

WORKED EXAMPLE 2 — THE STANDARD BOHR NUMERICALS

(a) **Radius of the 3rd orbit of Li^{2+} :** $Z = 3, n = 3$: $r = 52.9 \times \frac{9}{3} = \mathbf{158.7 \text{ pm}}$.

(b) **Ionisation energy of He^+ :** removing the electron means $n_1 = 1 \rightarrow n_2 = \infty$, so $\text{IE} = 13.6 \times Z^2 = 13.6 \times 4 = \mathbf{54.4 \text{ eV}}$.

(c) **Wavelength of the first Balmer line ($\text{H}_\alpha, n_2 = 3 \rightarrow n_1 = 2$):**

$\bar{\nu} = 109677 \left(\frac{1}{4} - \frac{1}{9} \right) = 109677 \times 0.1389 = 15,233 \text{ cm}^{-1} \Rightarrow \lambda = 1/15233 = 6.56 \times 10^{-5} \text{ cm} = \mathbf{656 \text{ nm (red)}}$.

(d) **Number of spectral lines** emitted when an electron drops from the n th level to the ground state $= \frac{n(n-1)}{2}$. From $n = 4$: 6 lines.

LIMITATIONS OF BOHR'S MODEL — A GUARANTEED QUESTION

1. Works **only for one-electron species**: H, He⁺, Li²⁺, Be³⁺. It fails for helium onward because it ignores electron–electron repulsion.
2. Cannot explain the **fine structure** of spectral lines revealed by better spectrometers.
3. Cannot explain the splitting of lines in a magnetic field (**Zeeman effect**) or an electric field (**Stark effect**).
4. Says nothing about the **shapes of molecules or chemical bonding**.
5. It violates the **Heisenberg uncertainty principle** by assigning a definite radius *and* a definite momentum at once — a fixed "orbit" is not allowed.

5 · Towards the Quantum Mechanical Model

5.1 de Broglie's dual nature of matter (1924)

DERIVATION 4 The de Broglie relation

1. For a photon, Planck gives $E = h\nu = \frac{hc}{\lambda}$, and Einstein's mass–energy relation gives $E = mc^2$.
2. Equating: $mc^2 = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{h}{mc} = \frac{h}{p}$.
3. de Broglie's leap: if radiation can behave as a particle, matter can behave as a wave. Replace c by the particle's speed v .

$$\lambda = \frac{h}{mv} = \frac{h}{p}. \text{ For an electron accelerated through } V \text{ volts, } \lambda = \frac{h}{\sqrt{2meV}} \approx \frac{12.27}{\sqrt{V}} \text{ \AA}.$$

Confirmed by **Davisson and Germer (1927)**, who diffracted electrons from a nickel crystal — diffraction is a purely wave phenomenon.

WHY WE NEVER SEE A CRICKET BALL DIFFRACT

λ is inversely proportional to mass. For a 100 g ball at 10 m s^{−1}, $\lambda \approx 6.6 \times 10^{-34}$ m — smaller than any aperture in the universe, so no diffraction is observable. Wave character only matters for particles of very small mass.

Bohr's fourth postulate now stops being arbitrary: a stable orbit must contain a whole number of electron wavelengths, $2\pi r = n\lambda = n\frac{h}{mv}$, which rearranges to exactly $mvr = \frac{nh}{2\pi}$.

5.2 Heisenberg's uncertainty principle (1927)

It is **impossible in principle** to determine simultaneously the exact position and the exact momentum of an electron: $\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$, equivalently $\Delta x \cdot \Delta v \geq \frac{h}{4\pi m}$.

This is not a limitation of instruments; it follows from the wave nature of matter. Its immediate casualty is the word "orbit": a precise path requires precise position and momentum together.

WORKED EXAMPLE 3 — WHY AN ELECTRON CANNOT RESIDE IN THE NUCLEUS

Confining an electron to a nucleus means $\Delta x \approx 10^{-14}$ m. Then

$$\Delta v \geq \frac{h}{4\pi m \Delta x} = \frac{6.626 \times 10^{-34}}{4\pi(9.11 \times 10^{-31})(10^{-14})} \approx 5.8 \times 10^9 \text{ m s}^{-1}$$

This exceeds the speed of light, which is impossible — so an electron cannot be confined inside the nucleus.

6 · The Quantum Mechanical Model of the Atom

Schrödinger's equation, $\hat{H}\psi = E\psi$, replaces orbits by **wave functions**. Solving it for the hydrogen atom yields the allowed energies E and the wave functions ψ automatically — the quantum numbers fall out of the mathematics rather than being assumed.

ψ itself has no physical meaning; $|\psi|^2$ is the **probability density** — the probability of finding the electron in a unit volume at that point. An **orbital** is the region of space where that probability is high (conventionally the surface enclosing 90% of the probability).

Orbit (Bohr)	Orbital (quantum mechanics)
A definite circular path	A region of probability; no path is defined
Position and momentum known together	Forbidden by the uncertainty principle
Always circular, two-dimensional	Different shapes — spherical, dumbbell, double dumbbell
Holds up to $2n^2$ electrons	Holds at most 2 electrons

6.1 Quantum numbers

Quantum number	Symbol	Allowed values	Tells you
Principal	n	1, 2, 3, ...	Shell; size and (mainly) energy; $2n^2$ electrons at most
Azimuthal (angular momentum)	l	0 to $(n - 1)$	Subshell and shape : $l = 0$ s, 1 p, 2 d, 3 f; orbital angular momentum = $\sqrt{l(l+1)} \frac{h}{2\pi}$
Magnetic	m_l	$-l \dots 0 \dots +l$ ($2l + 1$ values)	Orientation in space; the number of orbitals in the subshell
Spin	m_s	$+\frac{1}{2}$ or $-\frac{1}{2}$	Direction of electron spin; not a solution of the Schrödinger equation but required by experiment

Orbitals in a subshell = $2l + 1$ (s:1, p:3, d:5, f:7). Electrons in a subshell = $2(2l + 1)$ (s:2, p:6, d:10, f:14). Electrons in a shell = $2n^2$.

6.2 Shapes of orbitals and nodes

s orbitals are spherically symmetric, and the probability never vanishes in any direction. **p orbitals** are dumbbell-shaped, with the two lobes along the x , y or z axis and a **nodal plane** through the nucleus — the probability of finding a p electron *at* the nucleus is exactly zero. Four of the five **d orbitals** (d_{xy} , d_{yz} , d_{zx} , $d_{x^2-y^2}$) are four-lobed clover shapes; d_{z^2} has two lobes along z plus a doughnut in the xy plane.

Nodes are regions of zero probability. Angular (planar) nodes = l · Radial (spherical) nodes = $n - l - 1$ · **Total nodes** = $n - 1$.

WORKED EXAMPLE 4 — COUNTING NODES

For a **3p** orbital: $n = 3$, $l = 1$. Angular nodes = 1; radial nodes = $3 - 1 - 1 = 1$; total = $2 = n - 1$. ✓

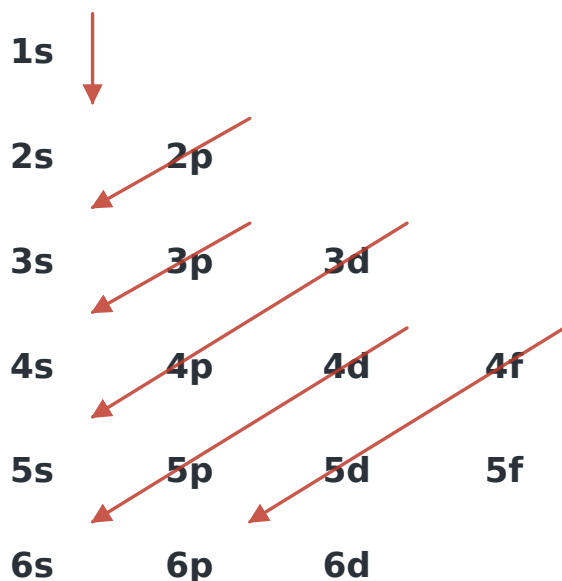
For a **3d** orbital: angular = 2, radial = $3 - 2 - 1 = 0$, total = 2. For a **4s**: angular = 0, radial = 3, total = 3.

6.3 Energies of orbitals

In **hydrogen** (one electron), energy depends on n *only*: $2s = 2p$, and $3s = 3p = 3d$. In **multi-electron atoms**, electron-electron repulsion splits them and the energy depends on **both** n and l .

Two competing effects explain the splitting. **Shielding**: inner electrons screen the nucleus, so an outer electron feels a reduced **effective nuclear charge** Z_{eff} . **Penetration**: within a shell, an s electron spends more time close to the nucleus than p, which does more than d, which does more than f. Greater penetration means less shielding, larger Z_{eff} and lower energy — hence **s < p < d < f** within a shell.

The $(n + l)$ rule. The orbital with the lower $(n + l)$ fills first; if two orbitals have the same $(n + l)$, the one with the **lower** n fills first. Example: 4s ($4 + 0 = 4$) fills before 3d ($3 + 2 = 5$); 3d ($= 5$) fills before 4p ($= 5$) because 3d has the smaller n .



Read along the arrows:

1s 2s 2p 3s 3p 4s 3d 4p

5s 4d 5p 6s 4f 5d 6p 7s

Equivalent to the $(n + l)$ rule:
lower $(n + l)$ fills first; on a tie,
the lower n fills first.

4s fills before 3d — but once
filled, 4s is also emptied first
on ionisation.

The Aufbau (building-up) order. Write the subshells in rows by shell and read along the diagonals. The 4s-before-3d crossover is the single most examined consequence.

6.4 Rules for filling electrons

1. **Aufbau principle.** Electrons occupy the lowest-energy orbital available first, in the $(n + l)$ order above.
2. **Pauli exclusion principle.** No two electrons in an atom can have all four quantum numbers the same — equivalently, an orbital holds at most two electrons and they must have **opposite spins**.
3. **Hund's rule of maximum multiplicity.** Pairing in the orbitals of a given subshell does not begin until each orbital of that subshell has one electron, and all the singly occupied orbitals have **parallel** spins.

6.5 Electronic configurations and the two famous exceptions

Configurations are written by subshell, e.g. O ($Z = 8$): $1s^2 2s^2 2p^4$, or in core notation [He] $2s^2 2p^4$.

Element	Expected	Actual
Chromium ($Z = 24$)	$[\text{Ar}] 3d^4 4s^2$	$[\text{Ar}] 3d^5 4s^1$
Copper ($Z = 29$)	$[\text{Ar}] 3d^9 4s^2$	$[\text{Ar}] 3d^{10} 4s^1$

Reason: exactly half-filled (d^5) and completely filled (d^{10}) subshells have extra stability, because of (i) the **symmetrical distribution** of electron density and (ii) the larger **exchange energy** released when electrons with parallel spins exchange positions — the more such exchanges possible, the more stable the arrangement.

WHERE THIS LOSES MARKS

1. **Electrons are removed from the outermost shell first, which means 4s before 3d** even though 4s was filled first. Fe ($Z = 26$) is $[\text{Ar}]3d^6 4s^2$, but Fe^{2+} is $[\text{Ar}]3d^6$ — *not* $[\text{Ar}]3d^4 4s^2$.
2. Configurations are **written** in shell order (3d before 4s) but **filled** in energy order (4s before 3d). Both conventions are accepted; be consistent.
3. Hund's rule is about a **subshell**, not the whole shell. $2p^3$ is three parallel unpaired electrons, one per orbital.

7 · Where marks are lost

THE TEN TRAPS OF CHAPTER 2

1. **Photoelectric KE depends on frequency; photocurrent depends on intensity.**
No amount of intensity below ν_0 ejects a single electron.
2. **Bohr's model applies only to one-electron species** — H, He^+ , Li^{2+} , Be^{3+} . Using $-13.6Z^2/n^2$ for lithium *atoms* is wrong.
3. **A spectral series is named by the level the electron lands on.** $n_1 = 2$ is always Balmer, whatever n_2 is.
4. **Energy is negative and rises towards zero** as n increases. "Higher energy" means *less negative*, and the gaps get smaller, never larger.
5. **Total nodes** = $n - 1$, of which l are angular. A 3s orbital has 2 radial nodes, not 3.
6. **In hydrogen 2s and 2p are degenerate; in every other atom they are not.**
Questions state which atom deliberately.
7. **m_l counts orientations, not electrons.** A p subshell has $2l + 1 = 3$ orbitals and therefore 6 electrons.
8. **Orbit vs orbital** — an orbital is a probability region and holds at most 2 electrons; only a Bohr orbit holds $2n^2$.
9. **Cr and Cu are the exceptions;** d^4s^2 and d^9s^2 do not occur. Quote symmetry *and* exchange energy as the reason.
10. **Isotopes differ in neutrons, isobars in protons.** "Same mass number, different atomic number" is an isobar — reversing the two is a standard one-mark loss.

Chapter 2 of the rationalised NCERT (2023-24 onward). Constants used: $h = 6.626 \times 10^{-34} \text{ J s}$, $c = 3.0 \times 10^8 \text{ m s}^{-1}$, $m_e = 9.11 \times 10^{-31} \text{ kg}$, $e = 1.602 \times 10^{-19} \text{ C}$, $R_H = 109,677 \text{ cm}^{-1}$.

Classification of Elements and Periodicity in Properties

Complete chapter notes — the genesis of the table, its structure from electronic configuration, every periodic trend with its reason, and the anomalies that examiners live on.

The spine of this chapter: two quantities explain almost every trend — the **effective nuclear charge** Z_{eff} pulling the outer electrons in, and the **number of shells** pushing them out. Across a period Z_{eff} rises with no new shell; down a group a new shell is added and shielding grows. Every trend in this chapter is one of those two effects winning. Anomalies happen only where a **stable half-filled or fully-filled subshell**, or an unusually **small size**, interferes.

1 · Genesis of Periodic Classification

Attempt	Idea	Why it failed
Döbereiner's triads (1829)	In a group of three similar elements, the atomic mass of the middle one is roughly the mean of the other two. Li (6.9), Na (23.0), K (39.1): mean = 23.0 ✓	Only a handful of triads could be found
Newlands' law of octaves (1865)	Arranged by increasing atomic mass, every eighth element repeats the properties of the first, like the notes of a musical scale	Valid only up to calcium; no room for the elements discovered later
Mendeleev's periodic law (1869)	"The properties of elements are a periodic function of their atomic masses ." Arranged in 8 groups and 6 periods, giving priority to chemical similarity over mass order	See the drawbacks below

1.1 Mendeleev — merits and drawbacks

Merits. (i) He left **gaps** for undiscovered elements and predicted their properties with remarkable accuracy — *eka*-aluminium (gallium), *eka*-boron (scandium) and *eka*-silicon (germanium) were all found later and matched. (ii) He **corrected wrong atomic masses** (beryllium, indium, gold) on the strength of chemical behaviour. (iii) Noble gases, discovered much later, slotted in as a new group without disturbing the table.

DRAWBACKS YOU MUST BE ABLE TO LIST

1. **Position of hydrogen** was uncertain — it resembles both the alkali metals ($1s^1$, forms H^+) and the halogens (one electron short of a full shell, forms H^- , diatomic).
2. **Anomalous pairs.** To keep chemically similar elements together, he had to place a heavier element before a lighter one: Ar (39.9) before K (39.1), Co (58.9) before Ni (58.7), Te (127.6) before I (126.9).
3. **No place for isotopes** — they have different masses but must occupy the same position.
4. Chemically dissimilar elements were grouped together (alkali metals with coinage metals Cu, Ag, Au).

1.2 Moseley and the modern periodic law

Moseley (1913) found that the frequency of the characteristic X-rays emitted by an element obeys $\sqrt{\nu} = a(Z - b)$ — a clean relationship with the **atomic number**, not the atomic mass. Atomic number is the fundamental property, and the anomalous pairs disappear the moment the table is ordered by Z .

Modern periodic law: the physical and chemical properties of the elements are a **periodic function of their atomic numbers**.

2 · Structure of the Modern Periodic Table

The long form has **18 vertical groups** and **7 horizontal periods**. Periodicity arises for one reason only: the **recurrence of similar outer-shell electronic configurations** after regular intervals.

	group →																	
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1																		
2																		
3																		
4																		
5																		
6			*															
7			**															

*																		
**																		

s-block: ns^{1-2} · p-block: ns^2np^{1-6} · d-block: $(n-1)d^{1-10}ns^{0-2}$ · f-block: $(n-2)f^{1-14}(n-1)d^{0-1}ns^2$

* lanthanoids (58–71, after La) · ** actinoids (90–103, after Ac) — taken out of period 6 and 7 to keep the t

Block structure of the periodic table. The block is named after the subshell that receives the last electron. Period length is fixed by which subshells are being filled, which is why the periods run 2, 8, 8, 18, 18, 32, 32.

Period	Subshells filled	Number of elements
1	1s	2
2	2s 2p	8
3	3s 3p	8
4	4s 3d 4p	18
5	5s 4d 5p	18
6	6s 4f 5d 6p	32
7	7s 5f 6d 7p	32

2.1 The four blocks

Block	Groups	General outer configuration	Characteristics
s	1, 2 (and He)	ns^{1-2}	Soft, highly reactive metals; low IE; strong reducing agents; lose electrons to form +1/+2 ions
p	13–18	ns^2np^{1-6}	Metals, non-metals and metalloids together; group 18 are the noble gases with filled shells
d (transition)	3–12	$(n-1)d^{1-10}ns^{0-2}$	Hard, high-melting metals; variable oxidation states; coloured ions; catalytic and magnetic behaviour
f (inner transition)	Placed separately	$(n-2)f^{1-14}(n-1)d^{0-1}ns^2$	Lanthanoids and actinoids; all metals; actinoids are radioactive

FREQUENTLY ASKED

Zinc, cadmium and mercury (group 12) have a completely filled d subshell in both the atom and all their common ions (d^{10}), so they are **not regarded as typical transition elements**. Helium is placed in group 18 despite the $1s^2$ configuration because its *properties* are those of a noble gas.

2.2 Locating an element from its configuration

Period number = the highest principal quantum number n in the configuration. **Group number:** s-block = number of valence electrons; p-block = $10 +$ number of electrons in $ns + np$; d-block = number of electrons in $(n-1)d + ns$.

WORKED EXAMPLE 1 — POSITION FROM CONFIGURATION

(a) $Z = 17$: $1s^2 2s^2 2p^6 3s^2 3p^5$. Highest $n = 3 \Rightarrow$ period 3; p-block with $2 + 5 = 7$ outer electrons $\Rightarrow$ group $10 + 7 =$ **17, period 3 (chlorine)**.

(b) $Z = 25$: $[\text{Ar}]3d^5 4s^2$. Highest $n = 4 \Rightarrow$ period 4; d-block with $5 + 2 = 7 \Rightarrow$ **group 7, period 4 (manganese)**.

(c) $Z = 20$: $[\text{Ar}]4s^2 \Rightarrow$ **group 2, period 4 (calcium)**.

2.3 Nomenclature of elements with $Z > 100$

Until an official name is agreed, IUPAC builds a temporary name directly from the digits of the atomic number, using numerical roots and the suffix **-ium**.

Digit	0	1	2	3	4	5	6	7	8	9
Root	nil	un	bi	tri	quad	pent	hex	sept	oct	enn

$Z = 104 \rightarrow \text{un} + \text{nil} + \text{quad} + \text{ium} = \text{unnilquadium (Unq)}$, now rutherfordium (Rf). $Z = 120 \rightarrow \text{unbinilium (Ubn)}$.

SPELLING RULES THAT CARRY A MARK

1. "bi" and "tri" are followed by **-ium**, giving "...bium", "...trium".
2. Drop one letter where the roots would collide: "enn" + "nil" $\rightarrow$ "ennil" (not "ennnil"); "bi"/"tri" + "ium" $\rightarrow$ "bium"/"trium".
3. The symbol is the **first letter of each root**, three letters in all.

3 · Periodic Trends in Physical Properties

3.1 Effective nuclear charge — the engine behind every trend

$Z_{\text{eff}} = Z - \sigma$, where σ is the **screening (shielding) constant** due to the inner electrons. **Across a period** Z rises by one each step while the added electron enters the *same* shell and screens poorly, so Z_{eff} **increases sharply**. **Down a group** both Z and the number of screening electrons rise, so Z_{eff} stays nearly constant while the number of shells increases.

Screening efficiency falls in the order **s > p > d > f**, because a more penetrating orbital keeps the electron closer to the nucleus.

3.2 Atomic radius

An atom has no sharp boundary, so radius is always defined operationally:

- **Covalent radius** — half the internuclear distance between two identical atoms bonded by a single covalent bond (Cl_2 : 198 pm apart $\Rightarrow$ 99 pm).
- **van der Waals radius** — half the distance between the nuclei of two *non-bonded* neighbouring atoms in the solid. Always **larger** than the covalent radius, because no orbital overlap is involved.
- **Metallic radius** — half the internuclear distance between adjacent atoms in a metallic crystal.

Across a period (left $\rightarrow$ right): radius **decreases** — Z_{eff} rises while the shell number stays the same. **Down a group**: radius **increases** — a new shell is added and shielding grows.

Period 2 (pm)	Li	Be	B	C	N	O	F
Atomic radius	152	111	88	77	74	66	64

WHERE THIS LOSES MARKS

1. **Noble gases are quoted as van der Waals radii**, so they look anomalously large at the end of a period. Never use them to argue that the radius rises at the right-hand edge.
2. Across the **d-block** the decrease is very gradual and then almost flat, because the added d electron shields the outer 4s electrons rather well.

3.3 Ionic radius

Cation < parent atom — an electron (often the whole outer shell) is removed while the nuclear charge is unchanged, so the remaining electrons are pulled in harder. **Anion > parent atom** — added electrons increase repulsion with no increase in nuclear charge.

Na (186 pm) $\rightarrow$ Na⁺ (95 pm); Cl (99 pm) $\rightarrow$ Cl⁻ (181 pm).

Isoelectronic species have the same number of electrons; among them, the radius **falls as the nuclear charge rises**:

N³⁻ (171) > O²⁻ (140) > F⁻ (136) > Na⁺ (95) > Mg²⁺ (65) > Al³⁺ (50) pm — all with 10 electrons.

MEMORY HOOK

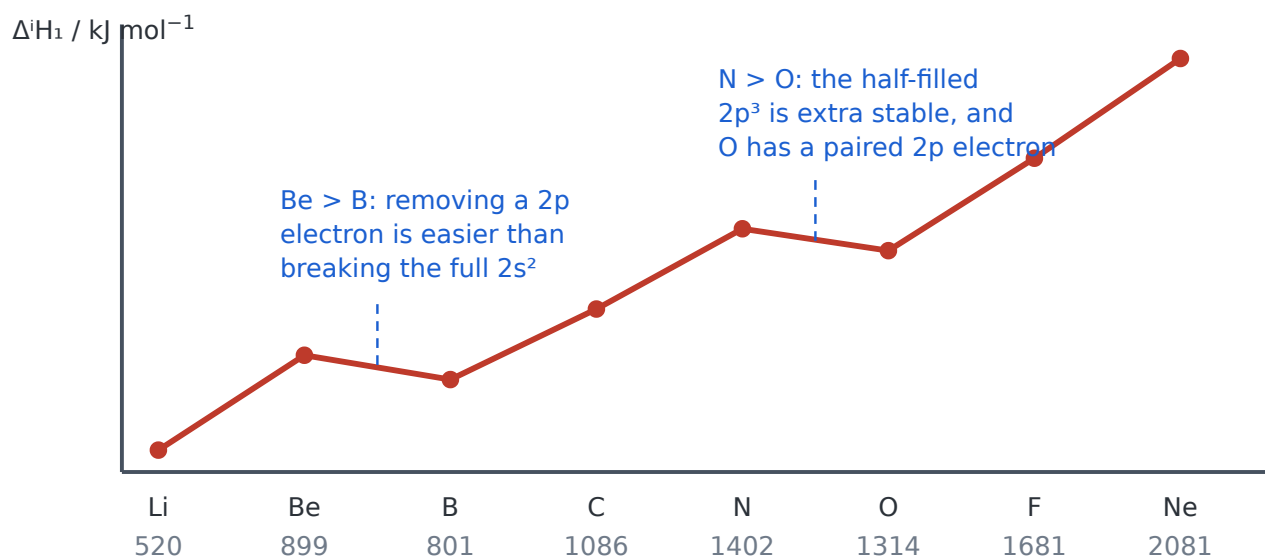
For an isoelectronic series, line the species up by **charge from most negative to most positive** — that order is already the order of decreasing size. More protons, same electrons, tighter grip.

3.4 Ionization enthalpy

$\Delta_i H_1$ is the energy required to remove the **most loosely bound electron** from an isolated **gaseous** atom in its ground state: $X(g) \rightarrow X^+(g) + e^-$. It is **always positive** (endothermic).

Factors: ionization enthalpy increases with larger Z_{eff} and decreases with larger atomic size and greater shielding; **stable half-filled and fully-filled** configurations raise it sharply.

Across a period: increases (size falls, Z_{eff} rises). **Down a group:** decreases (size rises, shielding rises). Highest in the period: the **noble gas**. Lowest: the **alkali metal**.



First ionization enthalpy across period 2. The overall rise is the Z_{eff} effect; the two dips are the whole examinable content of the graph. The identical pattern repeats in period 3 as **Mg > Al** and **P > S**.

The two anomalies, stated properly.

Be > B (and **Mg > Al**): boron's outgoing electron comes from a **2p** orbital, which is higher in energy and better shielded than beryllium's filled **2s²**, so less energy is needed.

N > O (and **P > S**): nitrogen has the **exactly half-filled 2p³** configuration, which is extra stable; oxygen's fourth 2p electron is **paired**, and the electron–electron repulsion within that orbital helps push it out.

3.5 Successive ionization enthalpies

$\Delta_i H_1 < \Delta_i H_2 < \Delta_i H_3 < \dots$ always. Removing an electron from an already positive ion is harder, because the remaining electrons are held by the same nuclear charge.

A **sudden jump** in the sequence signals that a noble-gas core has been reached, and reveals the number of valence electrons. Magnesium's third ionization enthalpy is enormous compared with its second, confirming that Mg has exactly two valence electrons and that Mg^{2+} is where it stops.

3.6 Electron gain enthalpy

$\Delta_{\text{eg}} H$ is the enthalpy change when an electron is added to an isolated gaseous atom: $\text{X(g)} + \text{e}^- \rightarrow \text{X}^-(\text{g})$. A **negative** value means energy is released — the standard convention, and the opposite in sign to "electron affinity".

$\Delta_{eg}H$ (kJ mol ⁻¹)	F	Cl	Br	I	O	S	Ne
Value	-328	-349	-325	-295	-141	-200	+116

Trends: becomes more negative across a period (halogens are the most negative of all) and less negative down a group. **Noble gases have large positive values** — the incoming electron must start a new shell.

THE CHLORINE-BEATS-FLUORINE ANOMALY — ASKED EVERY YEAR

$\Delta_{eg}H$ of **Cl is more negative than F**, and S more negative than O, even though F and O are higher in the group. **Reason:** the 2p subshell of fluorine is unusually **small and compact**, so the incoming electron suffers strong **inter-electronic repulsion** from the electrons already crowded there. Chlorine's larger 3p subshell accommodates the extra electron with far less repulsion.

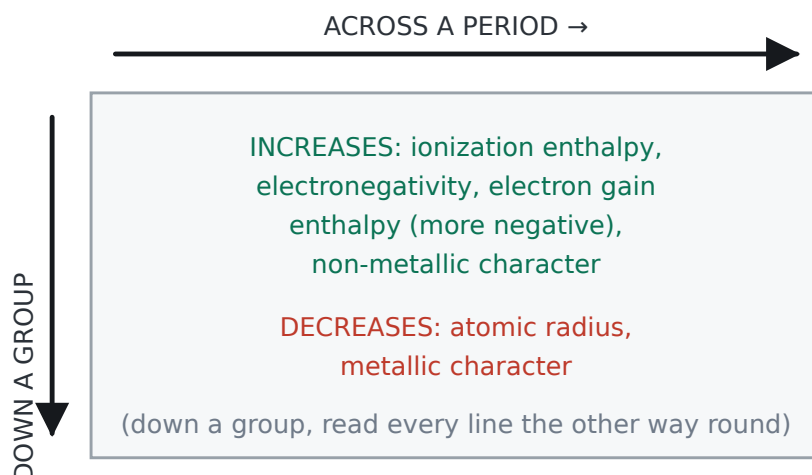
Also: the **second** electron gain enthalpy is always **positive** — forcing a second electron onto an already negative ion requires work. That is why O²⁻ formation is endothermic overall and only survives in a lattice.

3.7 Electronegativity

Electronegativity is the tendency of an atom **in a chemical compound** to attract the shared pair of electrons towards itself. It is a relative, unitless quantity — the Pauling scale sets F = 4.0, the highest of all elements.

	Electron gain enthalpy	Electronegativity
Applies to	An isolated gaseous atom	An atom within a bond
Nature	Measurable energy, in kJ mol ⁻¹	Relative number, no units

Trend: increases across a period (rising Z_{eff} , falling size) and decreases down a group. Order to remember: **F > O > N ≈ Cl > Br > C > H**. Metallic character runs exactly opposite to electronegativity.



Exceptions to memorise: Be > B, N > O, Mg > Al, P > S (ionization) · Cl > F, S > O (electron gain)

Trend compass. Everything except atomic radius and metallic character increases towards the top right of the table — which is why fluorine, at the top right of the reactive elements, is the most electronegative element there is.

4 · Periodic Trends in Chemical Properties

4.1 Valence and oxidation state

For representative elements the valence equals the number of valence electrons, or eight minus that number, whichever is smaller. Across period 3 the highest oxidation state with oxygen rises 1, 2, 3, 4, 5, 6, 7 (Na₂O, MgO, Al₂O₃, SiO₂, P₄O₁₀, SO₃, Cl₂O₇), because each of those electrons can be used in bonding.

4.2 Periodicity of oxides across period 3

Oxide	Na ₂ O	MgO	Al ₂ O ₃	SiO ₂	P ₄ O ₁₀	SO ₃	Cl ₂ O ₇
Nature	Basic		Amphoteric	Acidic (increasingly so)			

Amphoteric means reacting with both acids and bases: Al₂O₃ + 6HCl → 2AlCl₃ + 3H₂O, and Al₂O₃ + 2NaOH → 2NaAlO₂ + H₂O. Basic character falls and acidic character rises across a period; down a group the reverse happens (so, in group 15, N₂O₅ is strongly acidic while Bi₂O₃ is basic).

4.3 Anomalous behaviour of second-period elements

Three reasons, and the exam wants all three: (i) exceptionally **small size**; (ii) high **electronegativity** and high charge/radius ratio; (iii) **absence of d orbitals** in the valence shell, so the covalency cannot exceed **four**.

Consequence: nitrogen forms NF_3 but never NF_5 , while phosphorus forms both PF_3 and PF_5 using its 3d orbitals. Oxygen's maximum covalency is 2, sulfur's is 6 (SF_6).

4.4 Diagonal relationship

The first element of a group often resembles the **second element of the next group**, diagonally down-right: **Li-Mg, Be-Al, B-Si**.

Reason: moving right across a period decreases size and increases electronegativity, while moving down a group does the opposite. Along a diagonal the two effects nearly cancel, leaving similar size, charge/radius ratio and electronegativity. Hence LiCl and MgCl_2 are both deliquescent and soluble in ethanol; Li and Mg both form nitrides directly with N_2 ; BeO and Al_2O_3 are both amphoteric.

5 · Where marks are lost

THE TEN TRAPS OF CHAPTER 3

1. **The modern periodic law is about atomic number**, Mendeleev's was about atomic mass. Quote the right one for the right scientist.
2. **Ionization enthalpy is always positive; electron gain enthalpy is usually negative**. A "large" electron gain enthalpy means large in magnitude — say "more negative" to be unambiguous.
3. **Cl has a more negative $\Delta_{\text{eg}}H$ than F**, and S than O. The reason is small size and inter-electronic repulsion in the compact 2p subshell — not electronegativity.
4. **Be > B and N > O for ionization enthalpy**. Give the configuration reason ($2s^2$ filled; $2p^3$ half-filled), never just "exception".
5. **Second electron gain enthalpy is always positive**.
6. **Isoelectronic size falls as nuclear charge rises** — N^{3-} is the biggest and Al^{3+} the smallest of the 10-electron series.
7. **Noble-gas radii are van der Waals radii**, so they are not comparable with the covalent radii of their neighbours.
8. **Zn, Cd and Hg are not typical transition metals** (d^{10} in atom and ion alike).
9. **Electronegativity has no units and is not an energy**; electron gain enthalpy is measured in kJ mol^{-1} for a gaseous atom.
10. **Hydrogen has no fixed position** — it resembles both group 1 and group 17. Say so explicitly rather than assigning it to one.

Chapter 3 of the rationalised NCERT (2023–24 onward). All ionization and electron gain enthalpies quoted in kJ mol^{-1} ; radii in pm.

Chemical Bonding and Molecular Structure

Complete chapter notes — Lewis structures and formal charge, bond parameters, VSEPR shapes, hybridisation, molecular orbital theory worked for every diatomic, and hydrogen bonding.

The spine of this chapter: atoms bond to lower their energy, and there are exactly three theories, each answering a different question. **Lewis/VSEPR** answers *what shape?* **Valence bond theory (hybridisation)** answers *which orbitals overlap, and how strong is the bond?* **Molecular orbital theory** answers *how many bonds, and is it magnetic?* When a question mentions **paramagnetism or bond order**, it is asking for MOT; when it mentions **shape or bond angle**, it is asking for VSEPR.

1 • The Kössel-Lewis Approach

Atoms combine to attain the stable **octet** of the nearest noble gas — by transferring electrons (ionic bond) or by sharing them (covalent bond). A **Lewis symbol** shows the valence electrons as dots around the symbol; the number of dots gives the valence directly.

1.1 Writing a Lewis structure

METHOD 1 Drawing any Lewis structure

1. Count the **total valence electrons**. For an anion **add** one per unit of negative charge; for a cation **subtract** one per unit of positive charge.
2. Pick the **central atom** — the least electronegative one (never hydrogen, which forms only one bond).
3. Join each outer atom to the centre with a single bond (2 electrons each).
4. Distribute the remaining electrons as **lone pairs** on the outer atoms first, to complete their octets.
5. If the central atom is still short of an octet, convert lone pairs on the outer atoms into **multiple bonds**.
6. Check with **formal charges**; the best structure has formal charges closest to zero, with any negative charge on the most electronegative atom.

For CO_3^{2-} : $4 + 3(6) + 2 = 24$ valence electrons, C central, one C=O and two C-O⁻, giving three equivalent resonance structures.

Formal charge = (valence electrons in the free atom) – (lone-pair electrons) – $\frac{1}{2}$ (bonding electrons). Formal charges are a **bookkeeping device** for choosing between structures — they are not real charges on the atoms.

WORKED EXAMPLE 1 — FORMAL CHARGES IN OZONE

O₃ is drawn as O_{central} with one double and one single bond.

Central O: $6 - 2 - \frac{1}{2}(6) = +1$. Doubly bonded terminal O: $6 - 4 - \frac{1}{2}(4) = 0$. Singly bonded terminal O: $6 - 6 - \frac{1}{2}(2) = -1$.

Sum = $+1 + 0 - 1 = 0$, matching the neutral molecule — always check this.

1.2 Limitations of the octet rule

Failure	Explanation	Examples
Incomplete octet	Central atom has fewer than 8 electrons	LiCl, BeH ₂ , BCl ₃ (6 around B)
Expanded octet	Elements of period 3 onward have vacant d orbitals and may hold 10, 12 or more	PF ₅ (10), SF ₆ (12), H ₂ SO ₄ (12)
Odd-electron species	An odd total cannot be paired into octets	NO (11), NO ₂ (17)
Noble gas compounds	The octet was supposed to be inviolable	XeF ₂ , XeOF ₂ , KrF ₂

Beyond these, the rule says nothing about **molecular shape** and nothing about the **energy** (and hence the stability) of a molecule.

2 · The Ionic Bond

An ionic bond forms when one atom transfers electrons to another and the resulting oppositely charged ions are held by electrostatic attraction in a three-dimensional **lattice**.

Conditions that favour it: low ionization enthalpy of the metal, highly negative electron gain enthalpy of the non-metal, and — decisively — a **high lattice enthalpy**.

Lattice enthalpy is the energy required to completely separate one mole of a solid ionic compound into gaseous ions. For NaCl it is 788 kJ mol⁻¹. It rises with **higher ionic charges** and **smaller ionic radii** (roughly $\propto q_1q_2/r$), which is why MgO (2+, 2-) melts far above NaCl.

FREQUENTLY ASKED

The formation of Na^+ and Cl^- from the atoms is *endothermic* overall ($+496 - 349 = +147 \text{ kJ mol}^{-1}$). NaCl is stable only because the lattice enthalpy released on assembling the crystal far outweighs that cost. Ionic bonding is driven by the **lattice**, not by the ion formation.

2.1 Fajans' rules — covalent character in an ionic bond

Covalent character increases with (i) **small cation**, (ii) **large anion**, (iii) **high charge** on either ion, and (iv) a cation with a **pseudo-noble-gas (18-electron) configuration**, such as Cu^+ , Ag^+ , Zn^{2+} , which polarises better than an alkali-metal ion of the same size.

Hence LiCl is more covalent than NaCl ; AlCl_3 is more covalent than MgCl_2 ; and among the lithium halides, LiI is the most covalent. A more covalent halide is more soluble in organic solvents and has a lower melting point.

3 · Bond Parameters

Parameter	Definition	What to remember
Bond length	Equilibrium distance between the nuclei of two bonded atoms	Falls as bond order rises: $\text{C-C } 154 \text{ pm} > \text{C=C } 133 \text{ pm} > \text{C}\equiv\text{C } 120 \text{ pm}$
Bond angle	Angle between two bonds at an atom	Fixed by VSEPR; CH_4 109.5° , NH_3 107° , H_2O 104.5°
Bond enthalpy	Energy required to break one mole of bonds in the gaseous state	Rises with bond order: $\text{N}\equiv\text{N } 946 > \text{O}=\text{O } 498 > \text{H-H } 435.8 \text{ kJ mol}^{-1}$
Bond order	Number of bonds between two atoms	Higher order $\Rightarrow$ shorter bond, greater enthalpy, greater stability

For a polyatomic molecule such as water, breaking the two O-H bonds takes different energies (502 and 427 kJ mol^{-1}), so the tabulated value 464 is a **mean bond enthalpy**.

3.1 Resonance

When a single Lewis structure cannot describe a molecule adequately, the real structure is a **resonance hybrid** of several canonical forms. The hybrid is **lower in energy** — more stable — than any contributing structure.

In O_3 , both oxygen-oxygen bonds are **equal** at 128 pm — between a single bond (148 pm) and a double bond (121 pm). In CO_3^{2-} all three C-O bonds are identical, and in NO_3^- likewise, which no single Lewis structure can show.

THE CLASSIC MISCONCEPTION

1. The molecule does **not** flip between canonical forms, and it is **not** a mixture of them. It is a single structure that none of the drawings captures on its own.
2. Canonical forms have **no independent existence**; only the hybrid is real.
3. All canonical forms must have the **same positions of nuclei** — only electrons move. Moving an atom gives a different molecule, not a resonance form.

3.2 Polarity and dipole moment

$\mu = q \times d$, measured in **debye** ($1 \text{ D} = 3.33564 \times 10^{-30} \text{ C m}$). Dipole moment is a **vector**: the molecular dipole is the vector sum of the individual bond dipoles, so a molecule with polar bonds can still be non-polar.

Molecule	BeF ₂	CO ₂	BF ₃	CCl ₄	H ₂ O	NH ₃	NF ₃	HF
μ / D	0	0	0	0	1.85	1.47	0.24	1.78
Why	Symmetrical — bond dipoles cancel exactly				Unsymmetrical / lone pair adds to the dipole			

NH₃ VS NF₃ — THE SINGLE MOST ASKED COMPARISON

Both are pyramidal, yet $\mu(\text{NH}_3) = 1.47 \text{ D}$ while $\mu(\text{NF}_3)$ is only 0.24 D , even though N-F bonds are far more polar than N-H bonds.

Reason. In both molecules the nitrogen lone pair contributes a dipole pointing *away* from nitrogen, along the axis. In **NH₃** nitrogen is more electronegative than hydrogen, so the three bond dipoles point *towards* nitrogen — in the **same** direction as the lone-pair dipole, and the two **add**. In **NF₃** fluorine is more electronegative than nitrogen, so the bond dipoles point *away* from nitrogen — **opposing** the lone-pair moment, and they largely **cancel**.

4 · VSEPR Theory — Predicting Shape

Valence Shell Electron Pair Repulsion theory: the electron pairs in the valence shell of the central atom arrange themselves as far apart as possible, because like charges repel.

Repulsion order: lone pair-lone pair > lone pair-bond pair > bond pair-bond pair. A lone pair is held by only one nucleus, so it spreads out more and pushes harder.



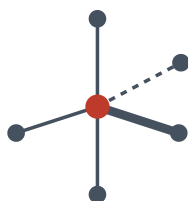
Linear · 180°
BeCl₂, CO₂



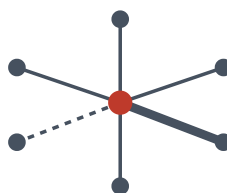
Trigonal planar · 120°
BF₃, BCl₃



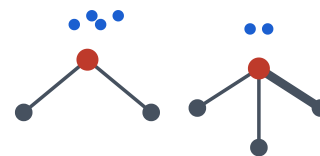
Tetrahedral · 109.5°
CH₄, NH₄⁺



Trigonal bipyramidal
PCl₅ · 90° and 120°



Octahedral · 90°
SF₆



Bent & pyramidal
H₂O 104.5° · NH₃ 107°

*The five basic geometries, plus the two commonest lone-pair shapes. Blue dots are lone pairs. The geometry is set by the total number of electron pairs; the **shape** is what you see when only the atoms are drawn.*

4.1 The complete VSEPR table

Type	Bond pairs	Lone pairs	Electron-pair geometry	Molecular shape	Example (angle)
AB ₂	2	0	Linear	Linear	BeCl ₂ , CO ₂ (180°)
AB ₃	3	0	Trigonal planar	Trigonal planar	BF ₃ (120°)
AB ₂ E	2	1	Trigonal planar	Bent	SO ₂ (119.5°), SnCl ₂
AB ₄	4	0	Tetrahedral	Tetrahedral	CH ₄ (109.5°)
AB ₃ E	3	1	Tetrahedral	Trigonal pyramidal	NH ₃ (107°)
AB ₂ E ₂	2	2	Tetrahedral	Bent	H ₂ O (104.5°)
AB ₅	5	0	Trigonal bipyramidal	Trigonal bipyramidal	PCl ₅ (90° , 120°)
AB ₄ E	4	1	Trigonal bipyramidal	See-saw	SF ₄
AB ₃ E ₂	3	2	Trigonal bipyramidal	T-shaped	ClF ₃
AB ₂ E ₃	2	3	Trigonal bipyramidal	Linear	XeF ₂ , I ₃ [−]
AB ₆	6	0	Octahedral	Octahedral	SF ₆ (90°)
AB ₅ E	5	1	Octahedral	Square pyramidal	BrF ₅
AB ₄ E ₂	4	2	Octahedral	Square planar	XeF ₄

MEMORY HOOK

In a **trigonal bipyramid**, lone pairs always take **equatorial** positions (only two neighbours at 90° instead of three), which is why SF_4 is a see-saw and ClF_3 is T-shaped. In an **octahedron** the second lone pair goes **trans** to the first, which is why XeF_4 is square planar rather than see-saw.

4.2 Bond-angle contraction

CH_4 (109.5°) > NH_3 (107°) > H_2O (104.5°). All three are sp^3 hybridised with four electron pairs; each lone pair added squeezes the bonding pairs closer together by about 2.5° .

Two further comparisons worth carrying: **NH_3 (107°) > NF_3 (102°)**, because the more electronegative fluorine pulls the bonding pairs away from nitrogen, reducing their mutual repulsion; and **NH_3 (107°) > PH_3 (93.6°)**, because the larger phosphorus atom uses nearly pure p orbitals, so the angle falls towards 90° .

5 · Valence Bond Theory and Hybridisation

VBT explains a covalent bond as the **overlap of two half-filled atomic orbitals** with opposite spins. Greater overlap gives a stronger bond, and the bond forms along the direction of maximum overlap — which is where directional character, and therefore shape, comes from.

5.1 Sigma and pi bonds

	σ (sigma) bond	π (pi) bond
Overlap	Head-on (axial): s-s, s-p, p-p along the axis	Sideways (lateral): p-p perpendicular to the axis
Strength	Stronger — overlap is larger	Weaker — overlap is smaller
Rotation	Free rotation possible about the bond	Restricted — this is the origin of geometrical isomerism
Occurrence	Always the first bond between two atoms	Only in addition to a σ bond

So a double bond is $1\sigma + 1\pi$, and a triple bond is $1\sigma + 2\pi$. In $\text{N}\equiv\text{N}$ (946 kJ mol^{-1}) the σ is formed by head-on $2p_z$ overlap and the two π bonds by sideways $2p_x$ and $2p_y$ overlap.

5.2 Hybridisation

Hybridisation is the intermixing of atomic orbitals of **nearly equal energy** belonging to the same atom, to give an equal number of **identical hybrid orbitals** of equal energy and shape. Hybrid orbitals are better directed and overlap more effectively, so they form stronger bonds.

Conditions: orbitals hybridised must belong to the same atom or ion, have comparable energies, and — importantly — hybridisation does *not* require the orbitals to be half filled; filled and empty orbitals may take part.

Hybridisation	Orbitals mixed	Geometry	Angle	s character	Examples
sp	1 s + 1 p	Linear	180°	50%	BeCl ₂ , C ₂ H ₂ , CO ₂
sp ²	1 s + 2 p	Trigonal planar	120°	33%	BF ₃ , C ₂ H ₄ , SO ₂
sp ³	1 s + 3 p	Tetrahedral	109.5°	25%	CH ₄ , NH ₃ , H ₂ O, NH ₄ ⁺
sp ³ d	1 s + 3 p + 1 d	Trigonal bipyramidal	90°, 120°	—	PCl ₅ , SF ₄ , ClF ₃
sp ³ d ²	1 s + 3 p + 2 d	Octahedral	90°	—	SF ₆ , XeF ₄

A CONSEQUENCE WORTH QUOTING

Greater **s character** means the hybrid orbital is held closer to the nucleus, so the bond is **shorter** and the atom behaves as more electronegative. This is why C-H bond length falls in the order sp³ (ethane, 109 pm) > sp² (ethene, 107 pm) > sp (ethyne, 106 pm), and why ethyne is acidic enough to react with sodium while ethane is not.

METHOD 2 Finding the hybridisation of any central atom

1. Compute the steric number $H = \frac{1}{2} [V + M - C + A]$, where V = valence electrons of the **central** atom, M = number of **monovalent** atoms attached (H, F, Cl...), C = charge if a cation, A = charge if an anion.
2. Read off: $H = 2 \rightarrow \text{sp}$, $3 \rightarrow \text{sp}^2$, $4 \rightarrow \text{sp}^3$, $5 \rightarrow \text{sp}^3\text{d}$, $6 \rightarrow \text{sp}^3\text{d}^2$.
3. Divalent atoms such as O and S attached to the centre contribute **nothing** to M — they take their electrons back through a double bond.

NH_4^+ : $\frac{1}{2}[5 + 4 - 1] = 4 \rightarrow \text{sp}^3$. SO_4^{2-} : $\frac{1}{2}[6 + 0 + 2] = 4 \rightarrow \text{sp}^3$. ClF_3 : $\frac{1}{2}[7 + 3] = 5 \rightarrow \text{sp}^3\text{d}$. XeF_4 : $\frac{1}{2}[8 + 4] = 6 \rightarrow \text{sp}^3\text{d}^2$. CO_2 : $\frac{1}{2}[4 + 0] = 2 \rightarrow \text{sp}$.

GEOMETRY IS NOT THE SAME AS SHAPE

The steric number gives the **electron-pair geometry**. The **shape** is what remains when the lone pairs are made invisible. H_2O is sp^3 with a *tetrahedral geometry* but a **bent shape**; XeF_4 is sp^3d^2 with an *octahedral geometry* but a **square planar shape**. Answer whichever the question asks for, and name the other to be safe.

6 · Molecular Orbital Theory

Atomic orbitals of the combining atoms merge into **molecular orbitals** that belong to the molecule as a whole. Two atomic orbitals combining by **LCAO** always give two molecular orbitals: one **bonding** (lower energy, additive combination, electron density between the nuclei) and one **antibonding** (higher energy, subtractive, with a node between the nuclei).

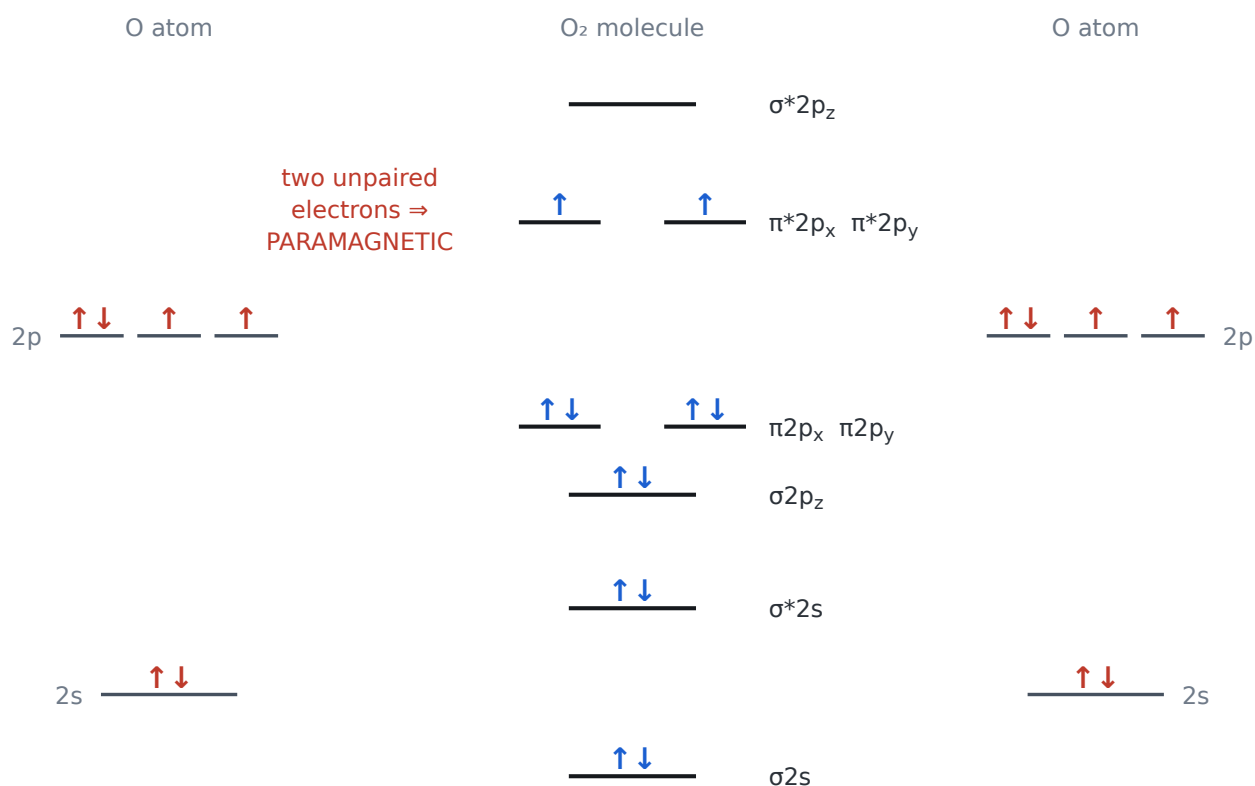
Conditions for combining: the atomic orbitals must have (i) **comparable energies**, (ii) the **same symmetry** about the molecular axis, and (iii) **sufficient overlap**.

Energy order.

For B_2 , C_2 , N_2 (**14 electrons or fewer**): $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \pi 2p_x = \pi 2p_y < \sigma 2p_z < \pi^* 2p_x = \pi^* 2p_y < \sigma^* 2p_z$

For O_2 , F_2 , Ne_2 (**more than 14**): the two swap — $\sigma 2p_z < \pi 2p_x = \pi 2p_y$.

Reason: in the lighter atoms the 2s and 2p levels are close enough to mix, and s-p mixing pushes $\sigma 2p_z$ above the π orbitals.



$N_b = 8, N_a = 4$ (valence orbitals) $\Rightarrow$ bond order $= (8 - 4)/2 = 2$, a double bond

MO diagram for O₂ (valence orbitals; the 1s electrons are written KK). The two unpaired electrons in the degenerate π* orbitals — placed there by Hund's rule — explain the **paramagnetism of oxygen**, which Lewis and valence bond theories both get wrong. For N₂ the π2p and σ2p levels are the other way round.

Bond order $= \frac{N_b - N_a}{2}$, where N_b and N_a are the numbers of bonding and antibonding electrons. A **positive** bond order means the molecule exists; **zero or negative** means it does not. Bond order $\uparrow \Rightarrow$ bond enthalpy $\uparrow$ and bond length $\downarrow$.

Species	Valence electrons	Bond order	Magnetic behaviour	Comment
H ₂	2	1	Diamagnetic	$\sigma 1s^2$
He ₂	4	0	—	Does not exist
Li ₂	2 (valence)	1	Diamagnetic	Exists in the vapour
Be ₂	4	0	—	Does not exist
B ₂	6	1	Paramagnetic	One electron in each of $\pi 2p_x, \pi 2p_y$
C ₂	8	2	Diamagnetic	Both bonds are π — no σ from 2p
N ₂	10	3	Diamagnetic	Shortest, strongest — 946 kJ mol^{-1}
O ₂	12	2	Paramagnetic	Two unpaired π^* electrons
F ₂	14	1	Diamagnetic	Weak single bond
Ne ₂	16	0	—	Does not exist

WORKED EXAMPLE 2 — THE OXYGEN FAMILY, A FAVOURITE ORDERING QUESTION

Adding electrons to O₂ puts them into **antibonding** π^* orbitals, so each one lowers the bond order:

Species	O ₂ ⁺	O ₂	O ₂ [−]	O ₂ ^{2−}
Valence electrons	11	12	13	14
Bond order	2.5	2	1.5	1
Unpaired electrons	1	2	1	0

Bond length: $\text{O}_2^+ < \text{O}_2 < \text{O}_2^- < \text{O}_2^{2-}$. **Stability** runs the other way. **O₂^{2−} is the only diamagnetic member.**

7 · Hydrogen Bonding

A **hydrogen bond** is the electrostatic attraction between a hydrogen atom covalently bonded to a highly electronegative, small atom (**F, O or N**) and a lone pair on another such atom. Strength $10\text{--}40 \text{ kJ mol}^{-1}$ — far weaker than a covalent bond (~ 400) but far stronger than ordinary van der Waals forces.

	Intermolecular	Intramolecular
Between	Two different molecules	Two groups within the same molecule
Effect	Associates molecules $\Rightarrow$ raises melting point, boiling point, viscosity and solubility in water	Ties the molecule up internally $\Rightarrow$ lowers boiling point and water solubility
Examples	H ₂ O, HF, NH ₃ , alcohols, carboxylic acid dimers, p -nitrophenol	o -nitrophenol, salicylaldehyde, salicylic acid

THE TWO ANSWERS THIS ALWAYS DEMANDS

Why is ice less dense than water? In ice each oxygen is tetrahedrally hydrogen-bonded to four others, producing an **open cage-like structure** with large empty spaces. On melting, some of these hydrogen bonds break, the cage collapses and the molecules pack closer — so water at 273 K is denser than ice, and ice floats.

Why does o-nitrophenol boil lower than p-nitrophenol? The *ortho* isomer forms an **intramolecular** hydrogen bond (the groups are adjacent), so its molecules are not associated with one another and it is volatile. The *para* isomer can only form **intermolecular** hydrogen bonds, associating the molecules and raising the boiling point — and making it far more soluble in water.

THE SUBTLE ORDERING

Hydrogen bond **strength** follows electronegativity: **HF > H₂O > NH₃**. But boiling point follows **H₂O > HF > NH₃**, because each water molecule has **two** hydrogens to donate *and* **two** lone pairs to accept, so it forms about four hydrogen bonds, whereas HF has only one hydrogen to donate. Total hydrogen bonding beats individual strength.

8 · Where marks are lost

THE TWELVE TRAPS OF CHAPTER 4

1. **Geometry $\neq$ shape.** H_2O : tetrahedral geometry, bent shape. XeF_4 : octahedral geometry, square planar shape.
2. **Lone pairs go equatorial in a trigonal bipyramid** and *trans* in an octahedron. This is why SF_4 is a see-saw and XeF_4 is square planar.
3. **NF_3 has a smaller dipole moment than NH_3** — bond dipoles oppose the lone-pair dipole in NF_3 and reinforce it in NH_3 .
4. **A molecule with polar bonds can be non-polar** if it is symmetrical: CO_2 , BF_3 , CCl_4 .
5. **Resonance forms do not interconvert** and do not exist separately; nuclei stay put, only electrons are redistributed.
6. **O_2 is paramagnetic** and only MOT explains it. Do not answer this with a Lewis structure.
7. **The MO order changes at 14 electrons.** Use π below σ_{2p} for B_2 , C_2 , N_2 ; σ_{2p} below π for O_2 , F_2 .
8. **Adding electrons to O_2 weakens it**, because they enter antibonding orbitals.
9. **Formal charge is bookkeeping**, not a real charge, and the formal charges must sum to the ion's overall charge.
10. **Bond angle order $\text{NH}_3 > \text{NF}_3$ and $\text{NH}_3 > \text{PH}_3$** — electronegativity of the surrounding atom, and size of the central atom, respectively.
11. **Hybridisation does not need half-filled orbitals**, and the number of hybrid orbitals always equals the number of atomic orbitals mixed.
12. **An intramolecular hydrogen bond lowers the boiling point**; only the intermolecular kind raises it.

Chapter 4 of the rationalised NCERT (2023–24 onward). Bond enthalpies in kJ mol^{-1} , bond lengths in pm, dipole moments in debye.

Redox Reactions

Complete chapter notes — oxidation numbers, every type of redox change, both balancing methods worked step by step, redox titrations and the electrode processes behind them.

The spine of this chapter: a redox reaction is nothing but **electron bookkeeping**. Assign oxidation numbers, see which element goes **up** (oxidation, electrons lost) and which goes **down** (reduction, electrons gained), and every question in the chapter — identify, classify, balance, titrate, predict — is answered from that one table of numbers. Oxidation and reduction always occur **together** and in exactly equal electron counts; that equality *is* the balancing method.

1 · Two Concepts of Oxidation and Reduction

	Oxidation	Reduction
Classical	Addition of oxygen or of any electronegative element; removal of hydrogen or of any electropositive element	Removal of oxygen; addition of hydrogen
Electronic	Loss of electrons	Gain of electrons
Oxidation number	Increase in oxidation number	Decrease in oxidation number

MEMORY HOOK

OIL RIG — **O**xidation **I**s **L**oss, **R**eduction **I**s **G**ain (of electrons). And remember the agents are named for what they do to the *other* substance: the **oxidising agent is itself reduced**, the **reducing agent is itself oxidised**.

The electronic definition is the general one — it covers reactions with no oxygen at all. In $2\text{Na(s)} + \text{Cl}_2\text{(g)} \rightarrow 2\text{NaCl(s)}$, sodium loses electrons (oxidised, reducing agent) and chlorine gains them (reduced, oxidising agent).

2 · Oxidation Number

Oxidation number is the charge an atom *appears* to have in a species when all the bonds are treated as fully ionic — the shared electrons of each bond assigned entirely to the more electronegative atom.

2.1 The rules, in order of priority

1. An atom in its **free element** has oxidation number **0**: Na, O₂, P₄, S₈, Cl₂.
2. A **monatomic ion** has an oxidation number equal to its charge: Na⁺ is +1, S²⁻ is -2.
3. **Fluorine is always -1** — no exceptions, it is the most electronegative element.
4. **Group 1 metals are +1**, group 2 are +2, aluminium is +3 in all their compounds.
5. **Hydrogen is +1** with non-metals, but **-1 in metal hydrides** (NaH, CaH₂, LiAlH₄).
6. **Oxygen is -2**, except: **-1 in peroxides** (H₂O₂, Na₂O₂), **- $\frac{1}{2}$ in superoxides** (KO₂), and **positive with fluorine** (+2 in OF₂, +1 in O₂F₂).
7. The **sum** of the oxidation numbers equals 0 for a neutral molecule and equals the charge for an ion.

WORKED EXAMPLE 1 — COMPUTING OXIDATION NUMBERS

(a) **Cr in Cr₂O₇²⁻**: $2x + 7(-2) = -2 \Rightarrow 2x = 12 \Rightarrow x = \mathbf{+6}$

(b) **Mn in MnO₄⁻**: $x + 4(-2) = -1 \Rightarrow x = \mathbf{+7}$

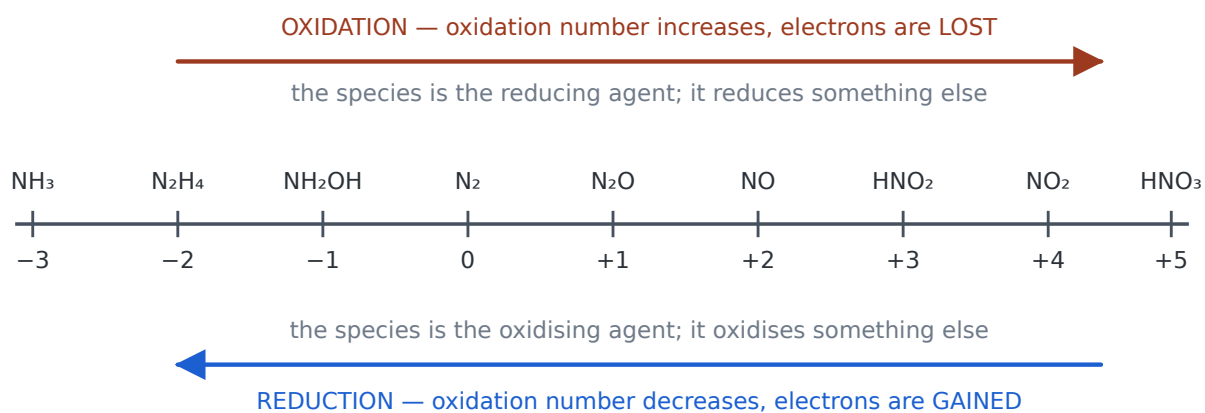
(c) **Fe in Fe₃O₄**: $3x + 4(-2) = 0 \Rightarrow x = \mathbf{+8/3}$ — a **fractional average**, because Fe₃O₄ is really FeO·Fe₂O₃, containing one Fe(II) and two Fe(III).

(d) **S in Na₂S₄O₆ (sodium tetrathionate)**: $2(+1) + 4x + 6(-2) = 0 \Rightarrow 4x = 10 \Rightarrow x = \mathbf{+2.5}$ — again an average: two of the sulfurs are at +5 and two, joined by an S-S bond, are at 0.

(e) **C in C₆H₁₂O₆**: $6x + 12(+1) + 6(-2) = 0 \Rightarrow x = \mathbf{0}$, even though the individual carbons are certainly not all zero.

OXIDATION NUMBER IS NOT VALENCY

1. Oxidation number may be **positive, negative, zero or fractional**; **valency is always a whole number and has no sign**. In CH₄ the oxidation number of carbon is -4 while its valency is 4.
2. A fractional value *always* means an **average** over atoms in genuinely different environments. Never write "the oxidation state of Fe is +8/3" without saying it is an average.
3. **Stock notation** puts the oxidation number in Roman numerals after the element: Fe(III) in Fe₂O₃, Mn(VII) in KMnO₄, Cu(I) in Cu₂O.



The oxidation-number line for nitrogen. Any species in the middle of its range can move both ways, and that is precisely the condition for **disproportionation**. An element already at its highest state (N in HNO_3) can only be reduced; at its lowest (N in NH_3) it can only be oxidised.

3 · Types of Redox Reactions

Type	Description	Example
Combination	Two species combine; redox only if an element is involved	$\text{C(s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$ $3\text{Mg} + \text{N}_2 \rightarrow \text{Mg}_3\text{N}_2$
Decomposition	A compound breaks into two or more products, at least one an element	$2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$ $2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2$
Displacement (metal)	A more reactive metal displaces a less reactive one from its salt	$\text{Zn(s)} + \text{CuSO}_4\text{(aq)} \rightarrow \text{ZnSO}_4\text{(aq)} + \text{Cu(s)}$
Displacement (non-metal)	Hydrogen or a halogen is displaced	$2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$ $\text{Cl}_2 + 2\text{KBr} \rightarrow 2\text{KCl} + \text{Br}_2$
Disproportionation	The same element in a single species is both oxidised and reduced	$2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$
Comproportionation	The reverse: two different oxidation states of one element give a single intermediate state	$5\text{Cl}^- + \text{ClO}_3^- + 6\text{H}^+ \rightarrow 3\text{Cl}_2 + 3\text{H}_2\text{O}$

3.1 Disproportionation — the type most often asked

Requirement: the element must be in an **intermediate** oxidation state, with both a higher and a lower state available to it.

Reaction	Element	Goes to
$2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$	O at -1	-2 (reduced) and 0 (oxidised)
$\text{Cl}_2 + 2\text{OH}^- \rightarrow \text{Cl}^- + \text{ClO}^- + \text{H}_2\text{O}$	Cl at 0	-1 and $+1$
$2\text{Cu}^+(\text{aq}) \rightarrow \text{Cu}^{2+}(\text{aq}) + \text{Cu}(\text{s})$	Cu at $+1$	$+2$ and 0
$\text{P}_4 + 3\text{OH}^- + 3\text{H}_2\text{O} \rightarrow \text{PH}_3 + 3\text{H}_2\text{PO}_2^-$	P at 0	-3 and $+1$

FREQUENTLY ASKED

Why does fluorine not disproportionate? Because fluorine is the most electronegative element and has **no positive oxidation state** — it can only be reduced from 0 to -1 , never oxidised. So F_2 with alkali gives OF_2 or O_2 (oxidising the *oxygen*), not the F^-/FO^- pair that chlorine gives.

4 · Balancing Redox Reactions

4.1 The oxidation number method

METHOD 1 Balancing by the oxidation number method

1. Write the skeletal equation and assign oxidation numbers to every atom.
2. Identify the elements whose oxidation numbers **change**, and compute the increase and the decrease **per atom**, then per formula unit.
3. Multiply the two species by whatever coefficients make the **total increase equal to the total decrease**.
4. Balance all remaining atoms except H and O by inspection.
5. Balance **oxygen** by adding H_2O , then **hydrogen** by adding H^+ (acidic medium).
6. In a **basic** medium, finish by adding as many OH^- ions to **both** sides as there are H^+ , and combine $\text{H}^+ + \text{OH}^-$ into H_2O .
7. Verify that both the **atoms** and the **total charge** balance.

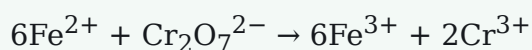
Step 3 is the whole method: electrons lost = electrons gained is the same statement as total increase = total decrease.

WORKED EXAMPLE 2 — DICHROMATE AND IRON(II) IN ACID

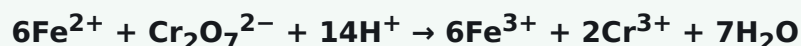
Skeleton: $\text{Fe}^{2+} + \text{Cr}_2\text{O}_7^{2-} + \text{H}^+ \rightarrow \text{Fe}^{3+} + \text{Cr}^{3+} + \text{H}_2\text{O}$

Changes: Fe goes $+2 \rightarrow +3$ — an increase of **1** per atom. Cr goes $+6 \rightarrow +3$ — a decrease of **3** per atom, and there are **2** Cr atoms, so **6** per dichromate ion.

Equalise: multiply Fe^{2+} by 6. $6 \times 1 = 1 \times 6$ ✓



Balance O with $7\text{H}_2\text{O}$ on the right, then H with 14H^+ on the left:



Charge check: left = $6(+2) + (-2) + 14(+1) = +24$; right = $6(+3) + 2(+3) = +24$.

Balanced.

4.2 The half-reaction (ion-electron) method

METHOD 2 Half-reaction method — acidic medium

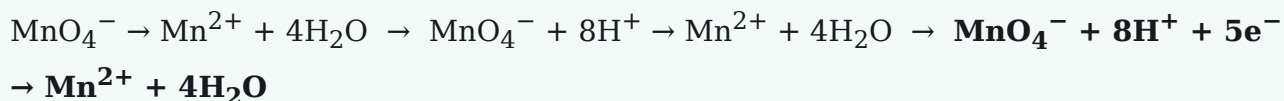
1. Split the reaction into the **oxidation half** and the **reduction half**.
2. In each half, balance all atoms **except H and O**.
3. Balance **O** by adding H_2O to the deficient side.
4. Balance **H** by adding H^+ .
5. Balance the **charge** by adding electrons — to the left of the reduction half, to the right of the oxidation half.
6. Multiply the two halves so that the **electrons cancel**, then add them together.
7. Cancel anything appearing on both sides, and check atoms and charge.

For a basic medium, do all of the above, then add OH^- to both sides equal to the number of H^+ , convert $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$, and cancel duplicated water molecules.

WORKED EXAMPLE 3 — PERMANGANATE AND SULFITE IN ACID

Skeleton: $\text{MnO}_4^- + \text{SO}_3^{2-} \rightarrow \text{Mn}^{2+} + \text{SO}_4^{2-}$ (acidic)

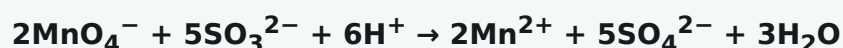
Reduction half (Mn: +7 $\rightarrow$ +2):



Oxidation half (S: +4 $\rightarrow$ +6):



Equalise electrons: reduction $\times 2$, oxidation $\times 5$ (LCM of 5 and 2 is 10), then add and cancel:

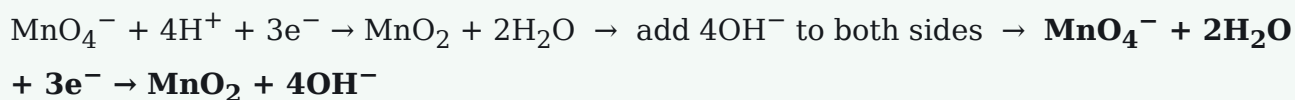


Check. O: $8 + 15 = 23$ left, $20 + 3 = 23$ right $\checkmark$. H: 6 both sides $\checkmark$. Charge: $-2 - 10 + 6 = -6$ left, $+4 - 10 = -6$ right $\checkmark$ **Balanced.**

WORKED EXAMPLE 4 — PERMANGANATE AND IODIDE IN ALKALI

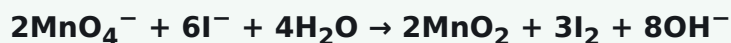
Skeleton: $\text{MnO}_4^- + \text{I}^- \rightarrow \text{MnO}_2 + \text{I}_2$ (basic)

Reduction half (Mn: +7 $\rightarrow$ +4), balanced first in acid then converted:



Oxidation half: $2\text{I}^- \rightarrow \text{I}_2 + 2\text{e}^-$

Multiply the first by 2 and the second by 3 (six electrons each), then add:



Charge check: left = $-2 - 6 = -8$; right = -8 $\checkmark$ **Balanced.**

WHERE BALANCING GOES WRONG

1. **Never add H^+ to a basic medium** in the final equation, and never add OH^- to an acidic one.
2. **Electrons must cancel completely.** If any remain in the final equation, the halves were not multiplied correctly.
3. **Check the total charge, not only the atoms.** A charge mismatch is the fastest way to catch an error.
4. Never change a formula (writing SO_4^- for SO_4^{2-}) to make the balancing work — only coefficients may change.

5 · Redox Titrations

The end point of a redox titration is found either from a **self-indicating** reagent ($KMnO_4$, whose first permanent pink tinge marks the end point) or from an **external indicator** (starch, giving a blue-black colour with iodine).

5.1 Permanganate — a different product in each medium

Medium	Half-reaction	Electrons (n -factor)	Mn goes
Acidic (dilute H_2SO_4)	$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$	5	+7 $\rightarrow$ +2 (colourless)
Neutral / faintly alkaline	$MnO_4^- + 2H_2O + 3e^- \rightarrow MnO_2 + 4OH^-$	3	+7 $\rightarrow$ +4 (brown solid)
Strongly alkaline	$MnO_4^- + e^- \rightarrow MnO_4^{2-}$	1	+7 $\rightarrow$ +6 (green)

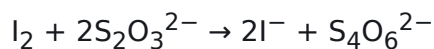
WHY SULFURIC ACID AND NEVER HCL OR HNO_3

Permanganate titrations are acidified with **dilute H_2SO_4** . Hydrochloric acid would itself be oxidised to chlorine, consuming extra $KMnO_4$ and giving a high result; nitric acid is itself an oxidising agent and would attack the reducing agent being estimated.

5.2 Dichromate and iodometry

Dichromate works only in acid, with $n = 6$: $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$. It is a **primary standard** — obtainable pure and stable — which $KMnO_4$ is not.

Iodometry liberates iodine, which is then titrated against standard sodium thiosulfate with starch near the end point:



Note the sulfur: +2 in thiosulfate rises to the average +2.5 in tetrathionate, so each thiosulfate loses exactly one electron.

5.3 Equivalents and n -factor

n -factor = number of electrons gained or lost per formula unit. Equivalent mass = $\frac{\text{molar mass}}{n\text{-factor}}$, Normality = Molarity $\times n$ -factor, and at the end point $N_1V_1 = N_2V_2$.

WORKED EXAMPLE 5 — A TITRATION CALCULATION

25.0 mL of an FeSO_4 solution needs 20.0 mL of 0.02 M KMnO_4 in dilute H_2SO_4 . Find the molarity of the iron(II) solution.

Reaction: $\text{MnO}_4^- + 5\text{Fe}^{2+} + 8\text{H}^+ \rightarrow \text{Mn}^{2+} + 5\text{Fe}^{3+} + 4\text{H}_2\text{O}$, so the mole ratio is **1 : 5**.

Moles of $\text{MnO}_4^- = 0.02 \times 0.0200 = 4.0 \times 10^{-4} \text{ mol} \Rightarrow \text{moles of Fe}^{2+} = 5 \times 4.0 \times 10^{-4} = 2.0 \times 10^{-3} \text{ mol}$

$$M(\text{Fe}^{2+}) = \frac{2.0 \times 10^{-3}}{0.0250} = \mathbf{0.080 \text{ M}}$$

Same answer by equivalents: $N(\text{KMnO}_4) = 0.02 \times 5 = 0.1 \text{ N}$; $N_1V_1 = N_2V_2 \Rightarrow N(\text{Fe}^{2+}) = \frac{0.1 \times 20}{25} = 0.08 \text{ N}$, and since $n = 1$ for Fe^{2+} , $M = 0.08 \text{ M}$.

6 · Redox Reactions and Electrode Processes

If the two halves of a redox reaction are physically separated, the electrons must travel through an external wire — and that current is usable work. This is a **galvanic (voltaic) cell**.

MEMORY HOOK

Activity series, most reactive first: **Li K Ba Ca Na Mg Al Zn Fe Ni Sn Pb (H) Cu Hg Ag Au** — "Little **K**atie **B**akes **C**akes, **N**aughty **M**aggie **A**lways **Z**aps **F**erocious **N**ice **S**nakes, **Pb**... **H**ere **C**urious **H**ungry **A**gents **A**udition." Any metal displaces every metal *below* it from solution.

7 · Where marks are lost

THE TWELVE TRAPS OF CHAPTER 7

1. **The oxidising agent is the one that gets reduced.** Write the oxidation numbers down before naming any agent.
2. **Oxidation number can be fractional; valency cannot.** A fraction always means an average over inequivalent atoms.
3. **Oxygen is not always -2** — it is -1 in peroxides, $-\frac{1}{2}$ in superoxides and **positive** in OF_2 . Fluorine, however, is always -1 .
4. **Hydrogen is -1 in metal hydrides** (NaH , CaH_2).
5. **Disproportionation needs an intermediate oxidation state.** Fluorine cannot disproportionate; nor can an element already at its highest or lowest state.
6. **Balance the charge as well as the atoms.** An equation with balanced atoms and unbalanced charge is still wrong.
7. **In basic medium, finish with OH^- , not H^+ .** Add OH^- to both sides and collapse the pairs into water.
8. **KMnO_4 gives a different product in each medium** — $n = 5$ in acid, 3 in neutral, 1 in strong alkali. Read the medium before writing the half-reaction.
9. **Acidify permanganate with dilute H_2SO_4 only**, never HCl (oxidised to Cl_2) or HNO_3 (itself an oxidant).
10. **Not every reaction with oxygen is redox, and not every redox has oxygen.** Precipitation and acid-base neutralisation involve no change in oxidation number at all.
11. **In a galvanic cell the anode is negative** (the opposite of an electrolytic cell) and oxidation always occurs there.
12. **E° values are not multiplied by stoichiometric coefficients.** Doubling a half-reaction does not double its electrode potential.

Chapter 7 of the rationalised NCERT (2023-24 onward) — the opening chapter of Chemistry Part II. Standard electrode potentials quoted at 298 K against the standard hydrogen electrode.

Master Formula Sheet

Every formula, constant and standard value from all five chapters in one place.

Universal constants. $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$ · $h = 6.626 \times 10^{-34} \text{ J s}$ · $c = 3.0 \times 10^8 \text{ m s}^{-1}$ · $m_e = 9.11 \times 10^{-31} \text{ kg}$ · $e = 1.602 \times 10^{-19} \text{ C}$ · $R_H = 109,677 \text{ cm}^{-1}$ · $1 \text{ u} = 1.66 \times 10^{-24} \text{ g}$ · $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$ · $1 \text{ bar} = 10^5 \text{ Pa}$ · $1 \text{ atm} = 1.013 \times 10^5 \text{ Pa}$

Chapter 1 · Some Basic Concepts of Chemistry

1 · The mole — the three bridges

MASS ↔ MOLE

$$n = \frac{m}{M}$$

m in g, M in g mol^{-1}

PARTICLES ↔ MOLE

$$n = \frac{N}{N_A}$$

$N_A = 6.022 \times 10^{23}$

GAS VOLUME ↔ MOLE (STP)

$$n = \frac{V}{22.7 \text{ L}}$$

22.7 L at 273.15 K & **1 bar**; 22.4 L at **1 atm**

SOLUTION ↔ MOLE

$$n = M \times V_{(L)}$$

M = molarity

2 · Masses and formulae

AVERAGE ATOMIC MASS

$$\bar{A} = \sum (\text{fractional abundance} \times \text{isotopic mass})$$

MASS % OF AN ELEMENT

$$\% E = \frac{\text{mass of E in 1 mol}}{\text{molar mass}} \times 100$$

MOLECULAR FROM EMPIRICAL

$$\text{MF} = (\text{EF})_n, \quad n = \frac{M}{\text{EFM}}$$

LIMITING REAGENT TEST

$$\text{smallest } \frac{\text{moles}}{\text{coefficient}}$$

never compare raw masses

% yield = $\frac{\text{actual yield}}{\text{theoretical yield}} \times 100$, theoretical yield being what the limiting reagent predicts.

3 • Concentration of solutions

Term	Formula	Unit	Depends on T?
Mass percentage	$\frac{w_B}{w_A + w_B} \times 100$	%	No
Mole fraction	$x_B = \frac{n_B}{n_A + n_B}$; $x_A + x_B = 1$	—	No
Molarity	$M = \frac{n_B}{V_{\text{solution}}(\text{L})}$	mol L ⁻¹	Yes
Molality	$m = \frac{n_B}{w_A(\text{kg})}$	mol kg ⁻¹	No
ppm	$\frac{\text{mass of solute}}{\text{mass of solution}} \times 10^6$	ppm	No
Normality †	$N = M \times n\text{-factor}$	eq L ⁻¹	Yes

DILUTION

$$M_1 V_1 = M_2 V_2$$

MIXING TWO SOLUTIONS †

$$M_f = \frac{M_1 V_1 + M_2 V_2}{V_1 + V_2}$$

MOLARITY FROM MASS % AND DENSITY

$$M = \frac{10 \times (\text{mass \%}) \times d}{M_B}$$

d in g mL⁻¹

MOLARITY → MOLALITY †

$$m = \frac{1000 M}{1000 d - M M_B}$$

4 • Units, significant figures, temperature

$$T(\text{K}) = t(^{\circ}\text{C}) + 273.15 \quad \cdot \quad t(^{\circ}\text{F}) = \frac{9}{5}t(^{\circ}\text{C}) + 32 \quad \cdot \quad 1 \text{ L} = 1 \text{ dm}^3 = 10^3 \text{ cm}^3 = 10^{-3} \text{ m}^3 \quad \cdot \quad \text{density}$$

SI unit kg m⁻³

Significant figures: addition/subtraction keeps the fewest **decimal places**;
multiplication/division keeps the fewest **significant figures**.

Chapter 2 • Structure of Atom

1 • Sub-atomic particles

Quantity	Value	Quantity	Value
e/m_e (Thomson)	$1.758820 \times 10^{11} \text{ C kg}^{-1}$	Mass of proton	$1.67262 \times 10^{-27} \text{ kg}$
e (Millikan)	$1.602 \times 10^{-19} \text{ C}$	Mass of neutron	$1.67493 \times 10^{-27} \text{ kg}$
Mass of electron	$9.1094 \times 10^{-31} \text{ kg}$	Neutrons	$A - Z$

2 • Radiation and the quantum

WAVE RELATION

$$c = \nu \lambda \quad \cdot \quad \bar{\nu} = \frac{1}{\lambda}$$

PLANCK / PHOTON ENERGY

$$E = h\nu = \frac{hc}{\lambda} \quad \cdot \quad E = nh\nu$$

PHOTOELECTRIC EQUATION

$$h\nu = h\nu_0 + \frac{1}{2}m_e v^2$$

$$KE_{\max} = h(\nu - \nu_0) ; \text{work function } \phi = h\nu_0$$

STOPPING POTENTIAL †

$$eV_0 = h(\nu - \nu_0)$$

EV SHORTCUT †

$$E(\text{eV}) = \frac{1240}{\lambda(\text{nm})}$$

RYDBERG FORMULA

$$\bar{\nu} = 109677 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ cm}^{-1}$$

Series	n_1	Region	Series	n_1	Region
Lyman	1	UV	Brackett	4	IR
Balmer	2	Visible	Pfund	5	Far IR
Paschen	3	IR	Lines to ground state = $\frac{n(n-1)}{2}$ †		

3 • Bohr's model — results and the derivation chain**RADIUS OF NTH ORBIT**

$$r_n = 52.9 \frac{n^2}{Z} \text{ pm}$$

$$r \propto n^2, r \propto 1/Z$$

ENERGY OF NTH ORBIT

$$E_n = -13.6 \frac{Z^2}{n^2} \text{ eV}$$

$$= -2.18 \times 10^{-18} \frac{Z^2}{n^2} \text{ J}$$

QUANTISED ANGULAR MOMENTUM

$$mvr = \frac{nh}{2\pi}$$

VELOCITY IN NTH ORBIT †

$$v_n = 2.188 \times 10^6 \frac{Z}{n} \text{ m s}^{-1}$$

FREQUENCY RULE

$$\Delta E = E_2 - E_1 = h\nu$$

ENERGY PARTITION †

$$PE = 2E \cdot KE = -E$$

DERIVATION CHAIN † Bohr radius and energy, in the order you write them

$$1. \text{Coulomb attraction} = \text{centripetal force: } \frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r^2} = \frac{mv^2}{r} \Rightarrow mv^2 = \frac{Ze^2}{4\pi\epsilon_0 r} \dots(i)$$

$$2. \text{Quantisation: } mvr = \frac{nh}{2\pi} \Rightarrow v = \frac{nh}{2\pi mr} \dots(ii)$$

$$3. (ii) \text{ into } (i): \frac{n^2 h^2}{4\pi^2 m r^2} = \frac{Ze^2}{4\pi\epsilon_0 r} \Rightarrow \boxed{r = \frac{n^2 h^2 \epsilon_0}{\pi m Z e^2}}$$

$$4. KE = \frac{1}{2}mv^2 = \frac{Ze^2}{8\pi\epsilon_0 r} ; PE = -\frac{Ze^2}{4\pi\epsilon_0 r}$$

$$5. E = KE + PE = -\frac{Ze^2}{8\pi\epsilon_0 r} \Rightarrow \boxed{E_n = -\frac{mZ^2 e^4}{8\epsilon_0^2 h^2 n^2}}$$

$$6. \text{Transition: } \Delta E = 2.18 \times 10^{-18} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ J} \Rightarrow \bar{\nu} = \frac{\Delta E}{hc} = 109677 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ cm}^{-1}$$

4 · de Broglie, Heisenberg, quantum numbers

DE BROGLIE WAVELENGTH

$$\lambda = \frac{h}{mv} = \frac{h}{p}$$

ACCELERATED ELECTRON †

$$\lambda = \frac{12.27}{\sqrt{V}} \text{ \AA}$$

UNCERTAINTY PRINCIPLE

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

$$\Delta x \cdot \Delta v \geq \frac{h}{4\pi m}$$

ORBITAL ANGULAR MOMENTUM

$$\sqrt{l(l+1)} \frac{h}{2\pi}$$

zero for every s orbital

Quantum number	Allowed values	Counting rule	Value
Principal n	1, 2, 3 ...	Electrons in a shell	$2n^2$
Azimuthal l	0 to $(n-1)$	Orbitals in a subshell	$2l+1$
Magnetic m_l	$-l \dots 0 \dots +l$	Electrons in a subshell	$2(2l+1)$
Spin m_s	$+\frac{1}{2}$ or $-\frac{1}{2}$	Filling order	lower $(n+l)$ first; tie $\rightarrow$ lower n

Nodes. Angular (planar) = l · Radial (spherical) = $n - l - 1$ · **Total** = $n - 1$ — the first two must add to the third.

Aufbau order: 1s 2s 2p 3s 3p 4s 3d 4p 5s 4d 5p 6s 4f 5d 6p 7s. **Exceptions:** Cr = [Ar]3d⁵4s¹, Cu = [Ar]3d¹⁰4s¹. **On ionisation 4s empties before 3d** (Fe²⁺ = [Ar]3d⁶).

Chapter 3 · Classification of Elements and Periodicity

Modern Periodic Law. The physical and chemical properties of the elements are periodic functions of their **atomic numbers**. **18 groups, 7 periods.** Period number = highest n in the configuration.

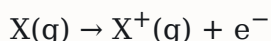
Elements per period: 2, 8, 8, 18, 18, 32, 32 · filled as 1s | 2s2p | 3s3p | 4s3d4p | 5s4d5p | 6s4f5d6p | 7s5f6d7p

Block	Groups	General configuration	Key features
s	1, 2	ns^{1-2}	Reactive metals, low IE, +1/+2, strong reducing agents
p	13-18	$ns^2 np^{1-6}$	Metals, non-metals and metalloids; group 18 noble gases
d	3-12	$(n-1)d^{1-10} ns^{0-2}$	Coloured ions, variable oxidation states, catalytic, paramagnetic. Zn/Cd/Hg excluded
f	separate	$(n-2)f^{1-14} (n-1)d^{0-1} ns^2$	Lanthanoids 58-71, actinoids 90-103 (radioactive); mainly +3

Trend summary — the whole chapter in one table

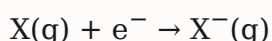
Property	Across a period →	Down a group ↓	Reason
Atomic radius	Decreases	Increases	Nuclear charge rises with no new shell / new shell added
Ionization enthalpy	Increases	Decreases	Size and shielding
Electron gain enthalpy	More negative	Less negative	Size
Electronegativity	Increases	Decreases	Size and nuclear charge
Metallic character	Decreases	Increases	Ease of losing electrons
Nature of oxide	Basic → amphoteric → acidic	More basic	—

IONIZATION ENTHALPY



always **positive**; $\Delta_i H_1 < \Delta_i H_2 < \Delta_i H_3$

ELECTRON GAIN ENTHALPY



negative if energy released; **2nd value always positive**

THE ANOMALIES — MEMORISE THESE FOUR LINES

1. **Be > B** and **Mg > Al** (ionization): the outgoing electron is from a higher-energy, better-shielded **p** orbital, versus breaking a filled **s²**.
2. **N > O** and **P > S** (ionization): half-filled **p³** is extra stable and symmetrical; the paired p electron in O suffers repulsion.
3. **Cl > F** and **S > O** (electron gain, more negative): F and O are so small that the compact 2p subshell causes strong **inter-electronic repulsion**.
4. **Isoelectronic radius** falls as nuclear charge rises: $O^{2-}(140) > F^-(136) > Na^+(95) > Mg^{2+}(65) > Al^{3+}(50)$ pm.

Period 2	Li	Be	B	C	N	O	F	Ne
Atomic radius / pm	152	111	88	77	74	66	64	—
$\Delta_i H_1$ / kJ mol ⁻¹	520	899	801	1086	1402	1314	1681	2080

Electron gain enthalpies (kJ mol^{-1}): F -328 , Cl -349 , Br -325 , I -295 , O -141 , S -200 . **Group 1 radii** (pm): Li 152, Na 186, K 231, Rb 244, Cs 262. **Electronegativity**: Pauling scale, F = 4.0 highest.

IUPAC roots ($Z > 100$): 0 nil · 1 un · 2 bi · 3 tri · 4 quad · 5 pent · 6 hex · 7 sept · 8 oct · 9 enn, then **-ium**. 104 = unnilquadium (Unq) · 120 = unbinilium (Ubn).

Second-period anomaly: small size, high charge/radius ratio and electronegativity, **no d orbitals** $\Rightarrow$ max covalency 4. **Diagonal pairs**: Li-Mg, Be-Al, B-Si.

Period 3 oxides: Na_2O , MgO (basic) · **Al_2O_3 (amphoteric)** · SiO_2 , P_4O_{10} , SO_3 , Cl_2O_7 (acidic). Neutral: CO, NO, N_2O .

Chapter 4 · Chemical Bonding and Molecular Structure

FORMAL CHARGE

$$\text{FC} = V - L - \frac{1}{2}B$$

valence – lone-pair electrons $-\frac{1}{2}(\text{bonding electrons})$

HYBRIDISATION / STERIC NUMBER [†]

$$H = \frac{1}{2} [V + M - C + A]$$

M = monovalent atoms; C cation charge; A anion charge

BOND ORDER (MOT)

$$\text{B.O.} = \frac{N_b - N_a}{2}$$

B.O. $\uparrow \Rightarrow$ bond enthalpy $\uparrow$, bond length $\downarrow$

DIPOLE MOMENT

$$\mu = q \times d$$

1 D = 3.336×10^{-30} C m; vector sum

Shapes — VSEPR geometry vs molecular shape



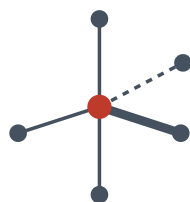
Linear · 180°
 BeCl_2 , CO_2



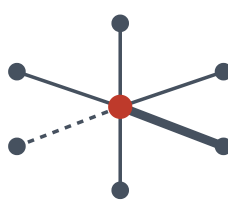
Trigonal planar · 120°
 BF_3 , BCl_3



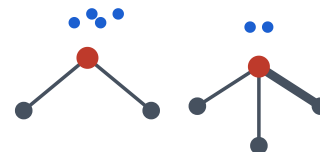
Tetrahedral · 109.5°
 CH_4 , NH_4^+



Trigonal bipyramidal
 PCl_5 · 90° and 120°



Octahedral · 90°
 SF_6



Bent & pyramidal
 H_2O 104.5° · NH_3 107°

The five basic geometries plus the two commonest lone-pair shapes. Blue dots are lone pairs. Geometry counts all electron pairs; shape is what remains when only atoms are drawn.

Type	BP	LP	Hybrid	Geometry	Shape	Example (angle)
AB ₂	2	0	sp	Linear	Linear	BeCl ₂ , CO ₂ (180°)
AB ₃	3	0	sp ²	Trigonal planar	Trigonal planar	BF ₃ (120°)
AB ₂ E	2	1	sp ²	Trigonal planar	Bent	SO ₂ (119.5°)
AB ₄	4	0	sp ³	Tetrahedral	Tetrahedral	CH ₄ (109.5°)
AB ₃ E	3	1	sp ³	Tetrahedral	Trigonal pyramidal	NH ₃ (107°)
AB ₂ E ₂	2	2	sp ³	Tetrahedral	Bent	H ₂ O (104.5°)
AB ₅	5	0	sp ³ d	Trigonal bipyramidal	Trigonal bipyramidal	PCl ₅ (90°, 120°)
AB ₄ E	4	1	sp ³ d	Trigonal bipyramidal	See-saw	SF ₄
AB ₃ E ₂	3	2	sp ³ d	Trigonal bipyramidal	T-shaped	ClF ₃
AB ₂ E ₃	2	3	sp ³ d	Trigonal bipyramidal	Linear	XeF ₂ , I ₃ ⁻
AB ₆	6	0	sp ³ d ²	Octahedral	Octahedral	SF ₆ (90°)
AB ₅ E	5	1	sp ³ d ²	Octahedral	Square pyramidal	BrF ₅
AB ₄ E ₂	4	2	sp ³ d ²	Octahedral	Square planar	XeF ₄

Repulsion order lp-lp > lp-bp > bp-bp. Lone pairs take **equatorial** positions in a trigonal bipyramid and go **trans** in an octahedron.

Bond angles: CH₄ 109.5° > NH₃ 107° > H₂O 104.5° · NH₃ 107° > NF₃ 102° · NH₃ 107° > PH₃ 93.6°

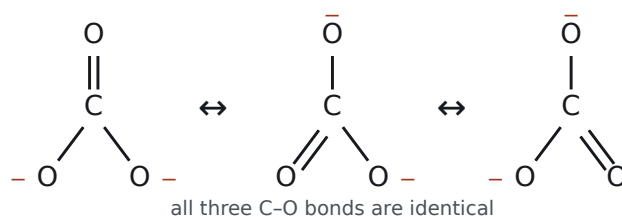
Dipole moments (D): BeF₂ 0 · CO₂ 0 · BF₃ 0 · CCl₄ 0 · **H₂O 1.85** · **NH₃ 1.47** · **NF₃ 0.24** · HF 1.78

Bond length / enthalpy: C-C 154 pm > C=C 133 > C≡C 120 · N≡N 946 > O=O 498 > H-H 435.8 kJ mol⁻¹ · lattice enthalpy NaCl 788 kJ mol⁻¹

Resonance structures



Ozone, O₃



Carbonate ion, CO₃²⁻

Rules. Canonical forms differ only in the **positions of electrons**, never of nuclei. They have **no independent existence** — only the hybrid is real, and it is **lower in energy** than any single form. All bonds in the hybrid are **equivalent**: O₃ both bonds 128 pm (between single 148 and double 121); CO₃²⁻ and NO₃⁻ all bonds identical — the nitrate ion has exactly the same three-form pattern as the carbonate ion drawn above.

Molecular orbital theory

Energy order for B₂, C₂, N₂ (≤ 14 electrons):



For O₂, F₂, Ne₂ (> 14 electrons): the last two swap — $\sigma 2p_z < \pi 2p_x = \pi 2p_y$

Species	H ₂	He ₂	Li ₂	Be ₂	B ₂	C ₂	N ₂	O ₂	F ₂	Ne ₂
Bond order	1	0	1	0	1	2	3	2	1	0
Magnetic	dia	—	dia	—	para	dia	dia	para	dia	—

Oxygen series	O ₂ ⁺	O ₂	O ₂ [−]	O ₂ ^{2−}
Valence electrons	11	12	13	14
Bond order	2.5	2	1.5	1
Unpaired electrons	1	2	1	0

Bond length O₂⁺ < O₂ < O₂[−] < O₂^{2−}; stability runs the other way. Added electrons enter **antibonding** orbitals.

Fajans' rules — covalent character rises with: small cation · large anion · high charge · pseudo-noble-gas (18-electron) cation.

Octet failures: incomplete (BCl₃, BeH₂) · expanded (PF₅, SF₆) · odd-electron (NO, NO₂) · noble gas compounds (XeF₂).

Hydrogen bond 10–40 kJ mol^{−1}, needs H bonded to **F, O or N**. Intermolecular **raises** boiling point; intramolecular (*o*-nitrophenol) **lowers** it. Strength HF > H₂O > NH₃, but boiling point H₂O > HF > NH₃.

Chapter 7 · Redox Reactions

Oxidation = loss of electrons = **increase** in oxidation number. **Reduction** = gain of electrons = **decrease** in oxidation number. **OIL RIG**.

The **oxidising agent is itself reduced**; the **reducing agent is itself oxidised**.

Oxidation number rules, in priority order

- Free element = **0** (Na, O₂, P₄, S₈, Cl₂)
- Monatomic ion = its charge (Na⁺ = +1, S^{2−} = −2)
- Fluorine is always** −1 — no exceptions
- Group 1 = +1, group 2 = +2, Al = +3
- Hydrogen = +1, but −1 **in metal hydrides** (NaH, CaH₂, LiAlH₄)
- Oxygen = −2, except −1 **in peroxides** (H₂O₂), $-\frac{1}{2}$ **in superoxides** (KO₂), **positive with F** (+2 in OF₂)

7. Sum = 0 for a neutral molecule, = the charge for an ion

Worked values: Cr in $\text{Cr}_2\text{O}_7^{2-} = +6$ · Mn in $\text{MnO}_4^- = +7$ · **Fe in $\text{Fe}_3\text{O}_4 = +8/3$**
(average) · **S in $\text{S}_4\text{O}_6^{2-} = +2.5$** (average) · C in $\text{C}_6\text{H}_{12}\text{O}_6 = 0$.

A **fractional** oxidation number always means an **average** over inequivalent atoms.
Oxidation number may be positive, negative, zero or fractional; **valency is always a whole number with no sign.**

Types of redox reaction

Type	Example
Combination	$\text{C(s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$ · $3\text{Mg} + \text{N}_2 \rightarrow \text{Mg}_3\text{N}_2$
Decomposition	$2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$ · $2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2$
Displacement (metal)	$\text{Zn(s)} + \text{CuSO}_4\text{(aq)} \rightarrow \text{ZnSO}_4\text{(aq)} + \text{Cu(s)}$
Displacement (non-metal)	$2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$ · $\text{Cl}_2 + 2\text{KBr} \rightarrow 2\text{KCl} + \text{Br}_2$
Disproportionation <i>same element both oxidised and reduced</i>	$2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$ (O: $-1 \rightarrow -2$ and 0) $\text{Cl}_2 + 2\text{OH}^- \rightarrow \text{Cl}^- + \text{ClO}^- + \text{H}_2\text{O}$ · $2\text{Cu}^+ \rightarrow \text{Cu}^{2+} + \text{Cu}$ $\text{P}_4 + 3\text{OH}^- + 3\text{H}_2\text{O} \rightarrow \text{PH}_3 + 3\text{H}_2\text{PO}_2^-$

DISPROPORTIONATION REQUIRES AN INTERMEDIATE OXIDATION STATE

Fluorine cannot disproportionate — it is the most electronegative element and has **no positive oxidation state**, so it can only be reduced from 0 to -1 .

Balancing — the two methods

METHOD 1 Oxidation number method

1. Assign oxidation numbers; identify which elements **change**.
2. Compute the increase and decrease **per atom**, then per formula unit.
3. Multiply so that **total increase = total decrease**.
4. Balance all other atoms except H and O by inspection.
5. Balance **O** with H_2O , then **H** with H^+ .
6. In basic medium, add OH^- to **both** sides equal to the H^+ , and combine into H_2O .
7. Check both **atoms and total charge**.

METHOD 2 Half-reaction (ion-electron) method

1. Split into the **oxidation half** and the **reduction half**.
2. Balance all atoms **except H and O**.
3. Balance **O** by adding H_2O .
4. Balance **H** by adding H^+ .
5. Balance **charge** by adding electrons.
6. Multiply so the **electrons cancel**, then add the halves.
7. **Basic medium**: add OH^- to both sides equal to the H^+ , convert $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$, cancel duplicated water.

Worked: $6\text{Fe}^{2+} + \text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ \rightarrow 6\text{Fe}^{3+} + 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$ (charge $+24 = +24$ ✓)

$2\text{MnO}_4^- + 5\text{SO}_3^{2-} + 6\text{H}^+ \rightarrow 2\text{Mn}^{2+} + 5\text{SO}_4^{2-} + 3\text{H}_2\text{O}$ (charge $-6 = -6$ ✓)

$2\text{MnO}_4^- + 6\text{I}^- + 4\text{H}_2\text{O} \rightarrow 2\text{MnO}_2 + 3\text{I}_2 + 8\text{OH}^-$ (basic; charge $-8 = -8$ ✓)

Standard half-reactions and n-factors

Reagent / medium	Half-reaction	n
KMnO₄ — acidic (dil. H_2SO_4)	$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$	5
KMnO ₄ — neutral / faintly alkaline	$\text{MnO}_4^- + 2\text{H}_2\text{O} + 3\text{e}^- \rightarrow \text{MnO}_2 + 4\text{OH}^-$	3
KMnO ₄ — strongly alkaline	$\text{MnO}_4^- + \text{e}^- \rightarrow \text{MnO}_4^{2-}$	1
K₂Cr₂O₇ — acidic only	$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$	6
Iodometry (thiosulfate)	$\text{I}_2 + 2\text{S}_2\text{O}_3^{2-} \rightarrow 2\text{I}^- + \text{S}_4\text{O}_6^{2-}$	1 per $\text{S}_2\text{O}_3^{2-}$

ACIDIFY PERMANGANATE WITH DILUTE H_2SO_4 ONLY

HCl would itself be oxidised to Cl_2 , consuming extra KMnO_4 ; HNO_3 is itself an oxidising agent and would attack the reducing agent being estimated.

Titration arithmetic and electrode processes

EQUIVALENT MASS

$$E = \frac{\text{molar mass}}{n\text{-factor}}$$

NORMALITY †

$$N = M \times n\text{-factor}$$

AT THE END POINT †

$$N_1 V_1 = N_2 V_2$$

CELL EMF

$$E_{\text{cell}}^\circ = E_{\text{cathode}}^\circ - E_{\text{anode}}^\circ$$

positive $\Rightarrow$ spontaneous

Cell notation: $\text{Zn(s)} \mid \text{Zn}^{2+}(1 \text{ M}) \parallel \text{Cu}^{2+}(1 \text{ M}) \mid \text{Cu(s)}$ — **anode left, cathode right**, double line = salt bridge.

Daniell cell: $E^\circ = 0.34 - (-0.76) = +1.10 \text{ V}$. Anode is **negative** in a galvanic cell and is where **oxidation** occurs. Electrons flow anode $\rightarrow$ cathode through the wire; the salt bridge carries ions the other way to keep both solutions neutral.

SHE: $\text{Pt} \mid \text{H}_2(1 \text{ bar}) \mid \text{H}^+(1 \text{ M})$, defined as exactly **0.00 V**. More **negative** E° = stronger **reducing** agent; more **positive** E° = stronger **oxidising** agent (F_2 , +2.87 V, is the strongest).

Activity series (most reactive first): Li K Ba Ca Na Mg Al Zn Fe Ni Sn Pb (H) Cu Hg Ag Au. Any metal displaces every metal **below** it from solution. Copper cannot liberate H_2 from dilute acid because its E° is positive.

E° values are never multiplied by stoichiometric coefficients.

CBSE Class 11 Chemistry, rationalised NCERT (2023–24 onward). Chapters 1–4 of Part I and Chapter 7 of Part II. Items marked [†] are standard exam formulae not stated in the NCERT text.